vm-grc-01 Locked-Down Azure VM: Evidence Binder

System owner: Tyrell Freeman. Evidence collected 2026-11-07 to 2026-11-10. Companion documents: Mini SSP and vm-grc-01_POAM_Risk_Register.xlsx.

Subscription IDs, billing IDs, email addresses, and phone numbers are redacted.

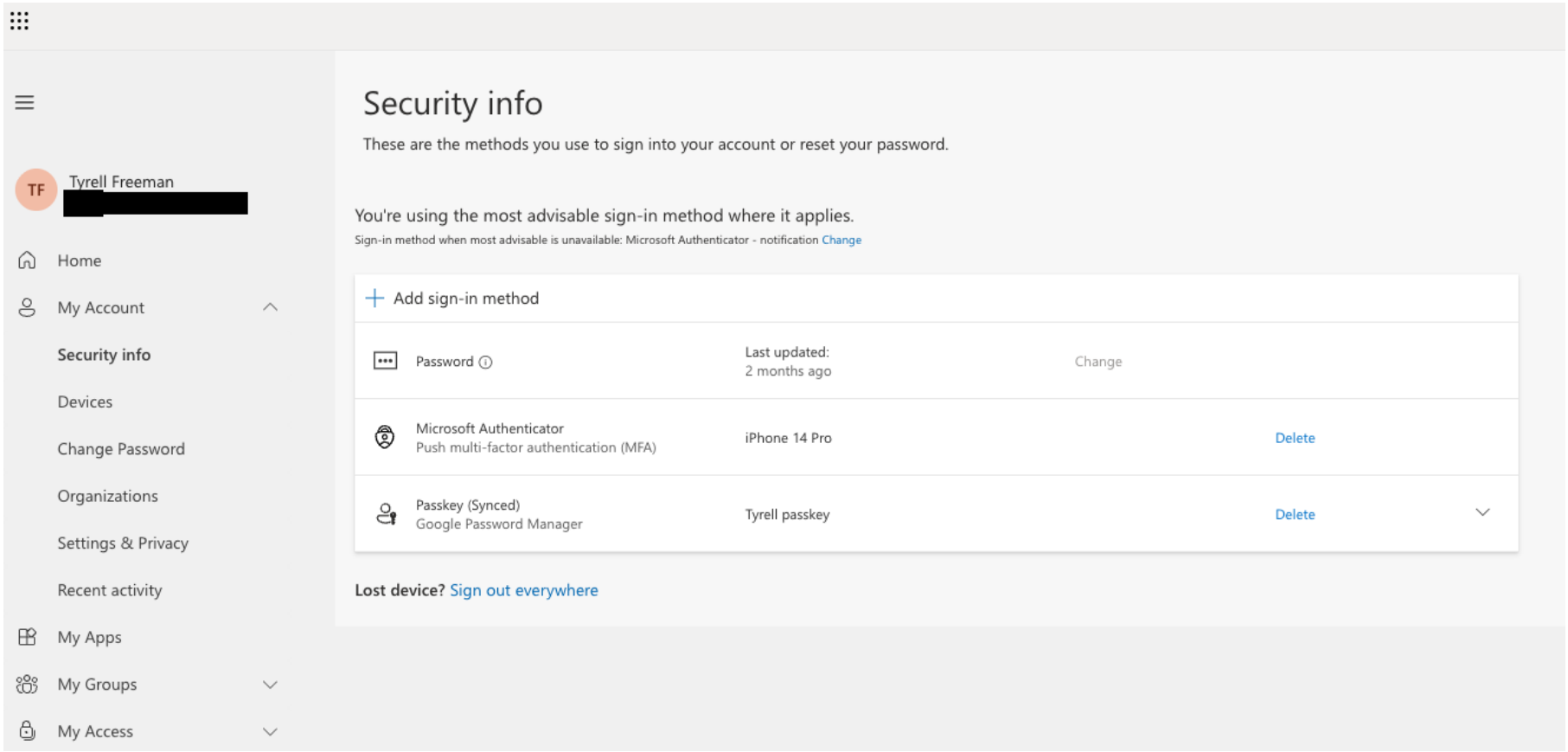
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ID	What it shows	Controls
Phase 0. Governance guardrails (Nov 7)		
E01	MFA registered on the St. John's Entra ID account (Authenticator push and passkey). Inherited control.	IA-2, IA-2(1)
E10a	Budget1: $20 per month at the billing account scope.	PM-3, SA-2
E10b	Actual-cost alerts at 50% ($10), 80% ($16), 100% ($20).	PM-3, SA-2
E10	Budget saved and active, $0 spent at creation.	PM-3, SA-2
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B01b	Tags owner, environment, project for inventory.	CM-8
B02	Subnet settings for the VM subnet (10.0.1.0/24).	SC-7
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B03c	Root cause: 'Allowed resource deployment regions' policy assignment.	CM-7, SA-9 (inherited)
B03d	Allowed regions parameter; West US 2 chosen.	CM-7
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B04	Final VNet vnet-grc-lab, 10.0.0.0/16, West US 2.	SC-7, CM-8
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B06	nsg-vm associated to subnet snet-vm1.	SC-7
E03	Inbound rules: defaults only, DenyAllInBound at 65500, no custom allow rules.	SC-7, AC-17
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B07b	SSH public key authentication; public inbound ports None.	IA-5, CM-7
E08	OS disk Standard SSD, encrypted at rest with platform-managed key.	SC-28
B08	Networking: snet-vm1, Public IP None, NIC NSG None (subnet NSG applies).	SC-7, AC-17

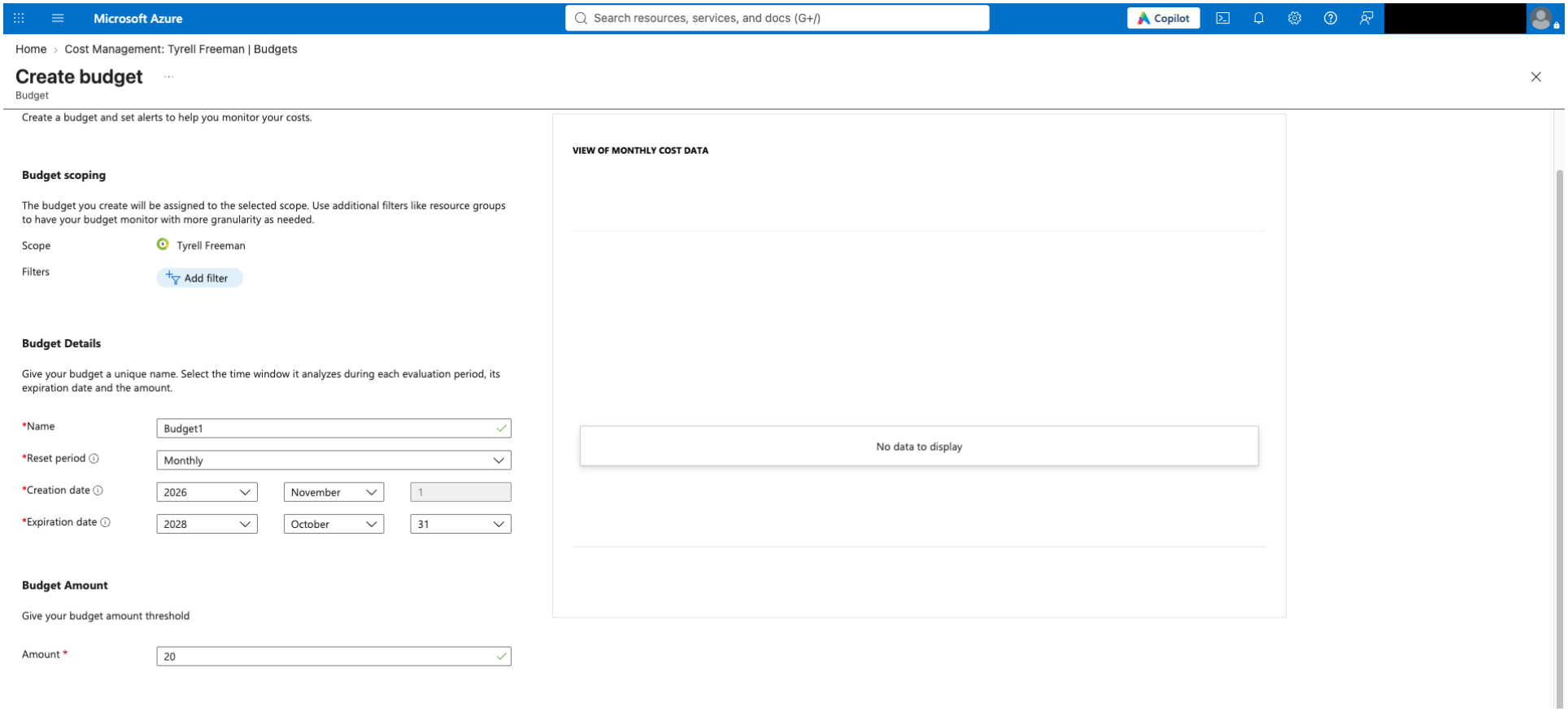
ID	What it shows	Controls
E09	Auto-shutdown 7:00 PM ET; periodic patch assessment on.	SI-2, PM-3
B09a	Final review: price, basics, Trusted launch settings.	CM-2
B09b	Final review: authentication, disks, networking.	CM-2
B09c	Final review: management and monitoring settings.	CM-2
B09d	Final review: tags applied to all resources.	CM-8
E02	VM running with no public IP; private IP 10.0.1.4 only.	SC-7, AC-17, CM-8
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E05b	SSH session established through Bastion to 10.0.1.4.	AC-17(3)
E06 / E07a	sshd: password auth off, key auth on. FINDING: apt said 'up to date' while every repository fetch timed out.	IA-5, SI-2
B11	NAT gateway deployed to give the private subnet an egress path.	SC-7
B12	Subnet confirmed private, with NAT gateway and nsg-vm attached.	SC-7
B13	Root cause: NAT gateway had no outbound public IP. Fixed by adding one.	SI-2 (root cause)
E07b	apt upgrade succeeds after the fix; updates installed.	SI-2
E07c	apt list --upgradable returns nothing: fully patched.	SI-2
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E11a	NIST SP 800-53 Rev. 5 initiative assigned to rg-grc-lab.	CA-2, CA-7
E11b	Baseline scan: 22 of 694 policies non-compliant (triaged in the POA&M; workbook).	CA-2, CA-7, RA-5
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F07	POAM-07 closed: root login disabled; /etc/passwd 644.	CM-6, AC-6
F01	POAM-01: Policy remediation tasks added managed identity and Guest Configuration extension.	CM-6
F02	POAM-02 closed: unused AzureBastionSubnet removed.	SC-7
F06	POAM-06: NIC IP forwarding disabled, no public IP.	SC-7, AC-4
F09	POAM-09 closed: second budget alert recipient added.	PM-3

Phase 0. Governance guardrails (Nov 7)

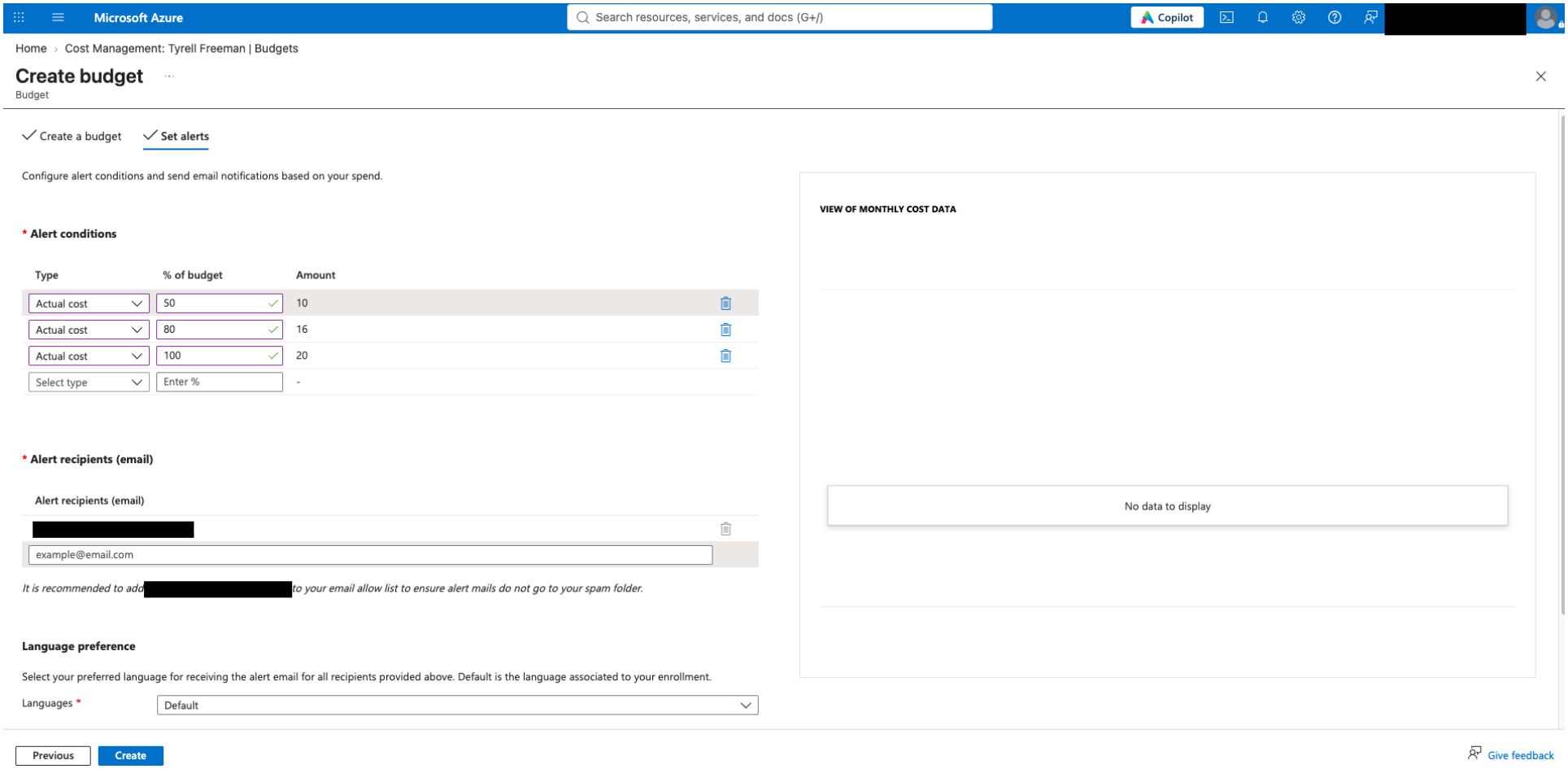
E01: MFA registered on the St. John's Entra ID account (Authenticator push and passkey). Inherited control. Controls: IA-2, IA-2(1)



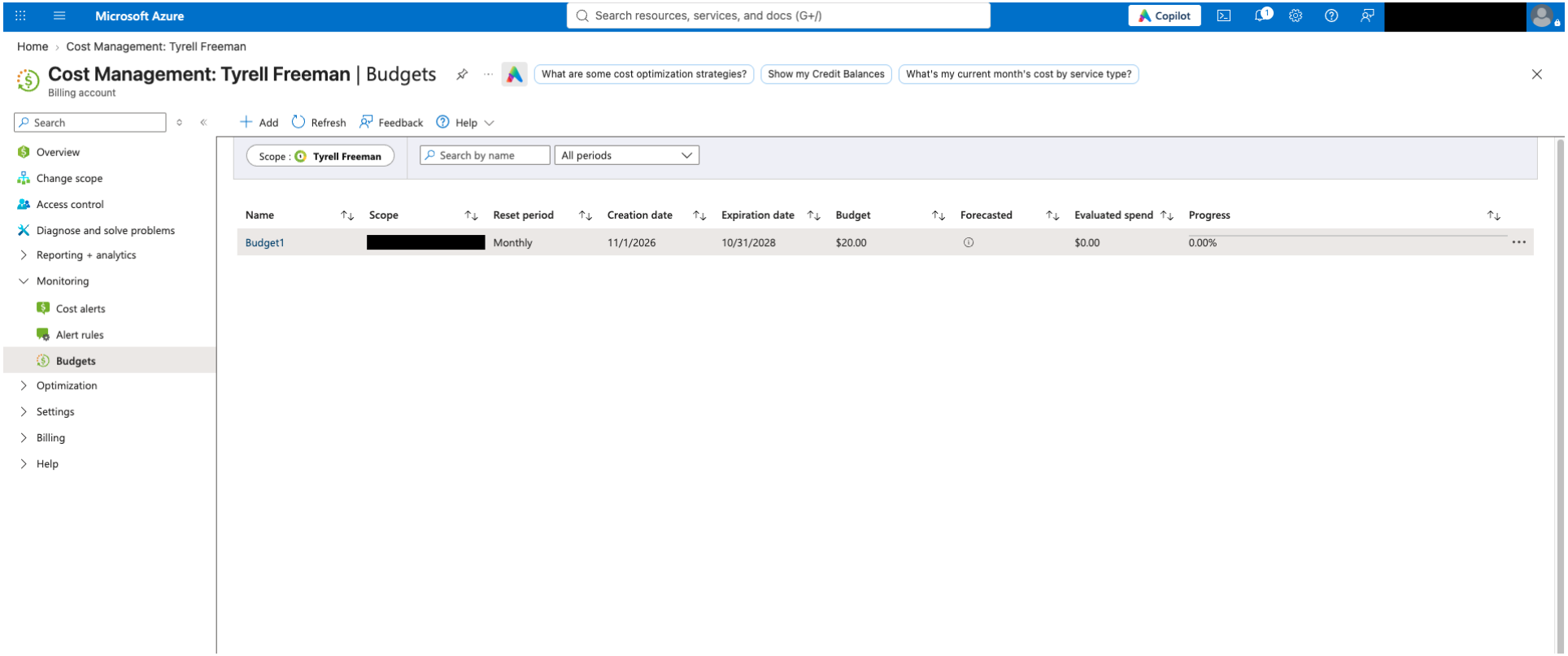
E10a: Budget1: $20 per month at the billing account scope. Controls: PM-3, SA-2



E10b: Actual-cost alerts at 50% ($10), 80% ($16), 100% ($20). Controls: PM-3, SA-2

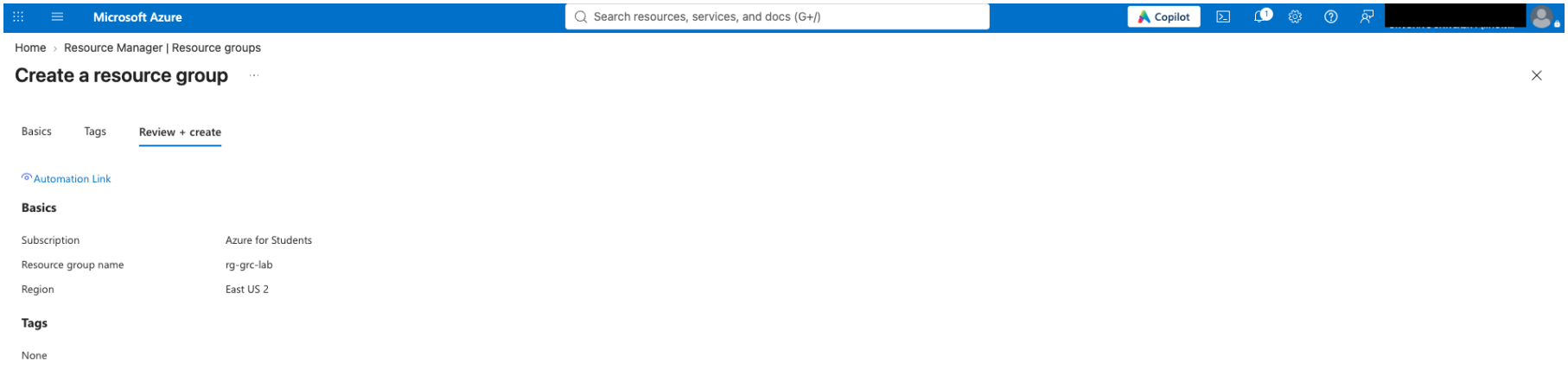


E10: Budget saved and active, $0 spent at creation. Controls: PM-3, SA-2

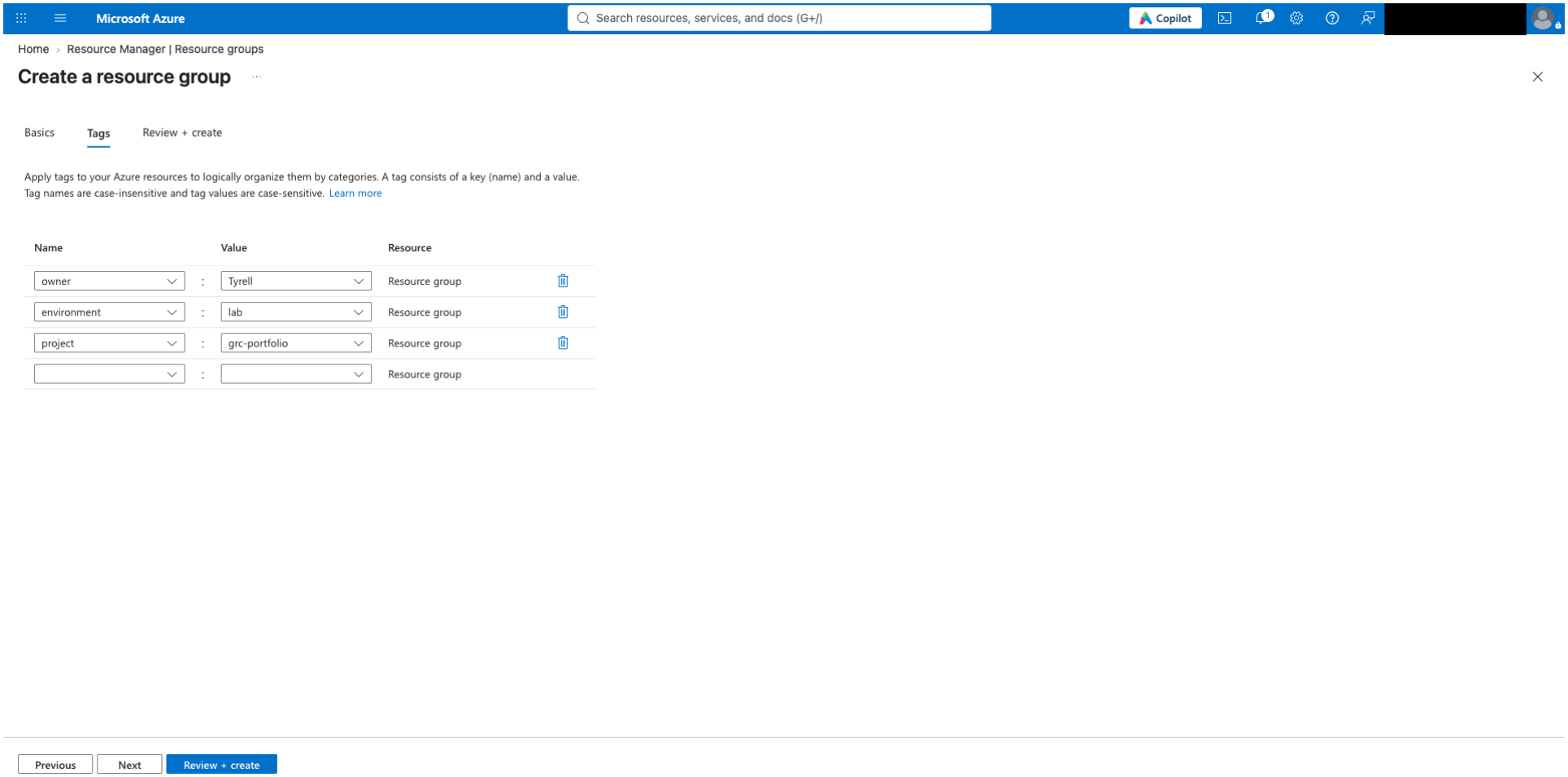


Phase 1. Network build (Nov 7)

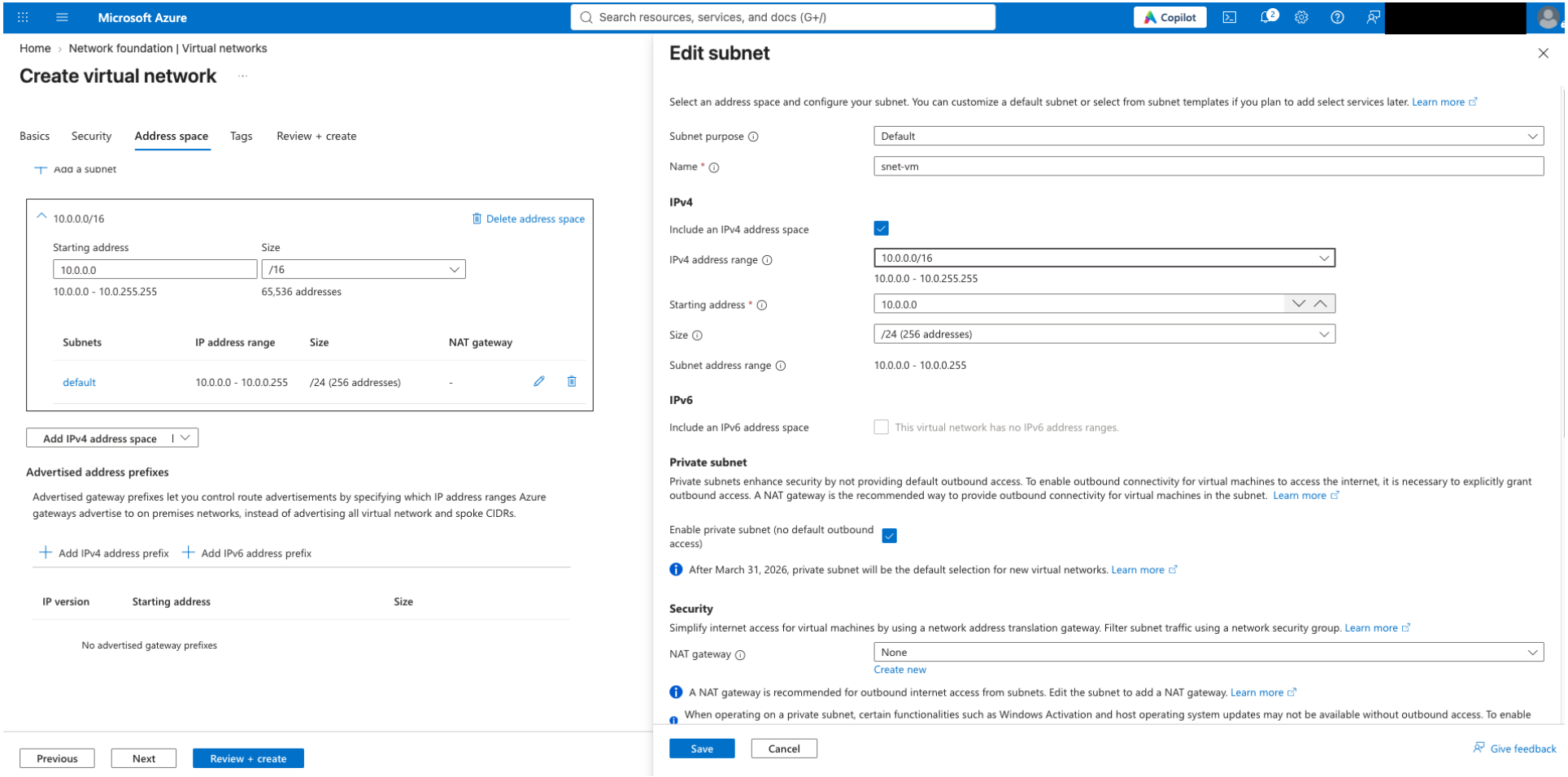
B01: Resource group rg-grc-lab created. Controls: CM-8



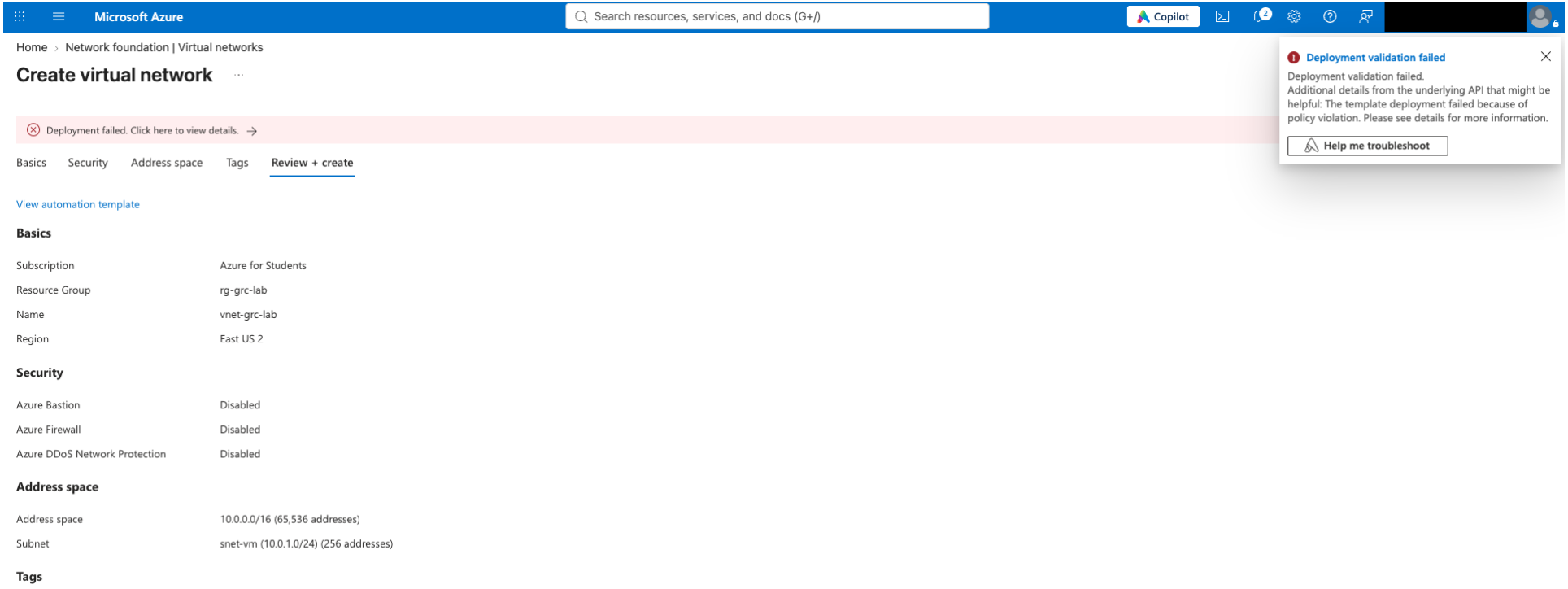
B01b: Tags owner, environment, project for inventory. Controls: CM-8



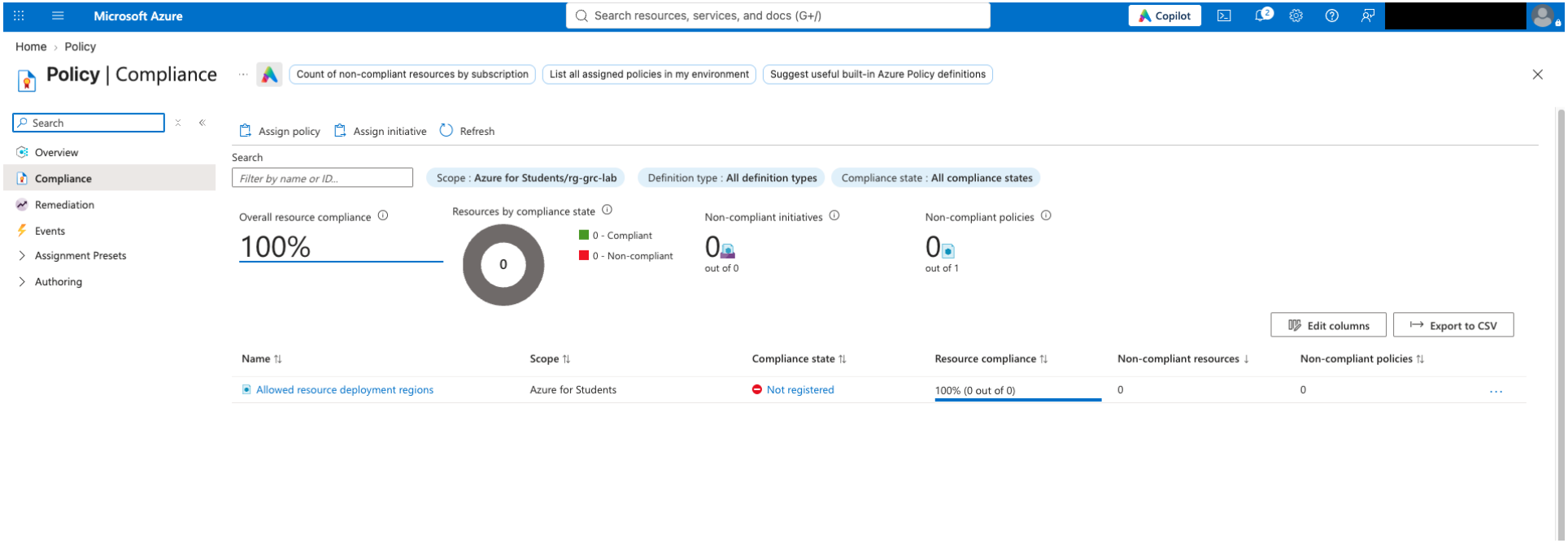
B02: Subnet settings for the VM subnet (10.0.1.0/24). Controls: SC-7



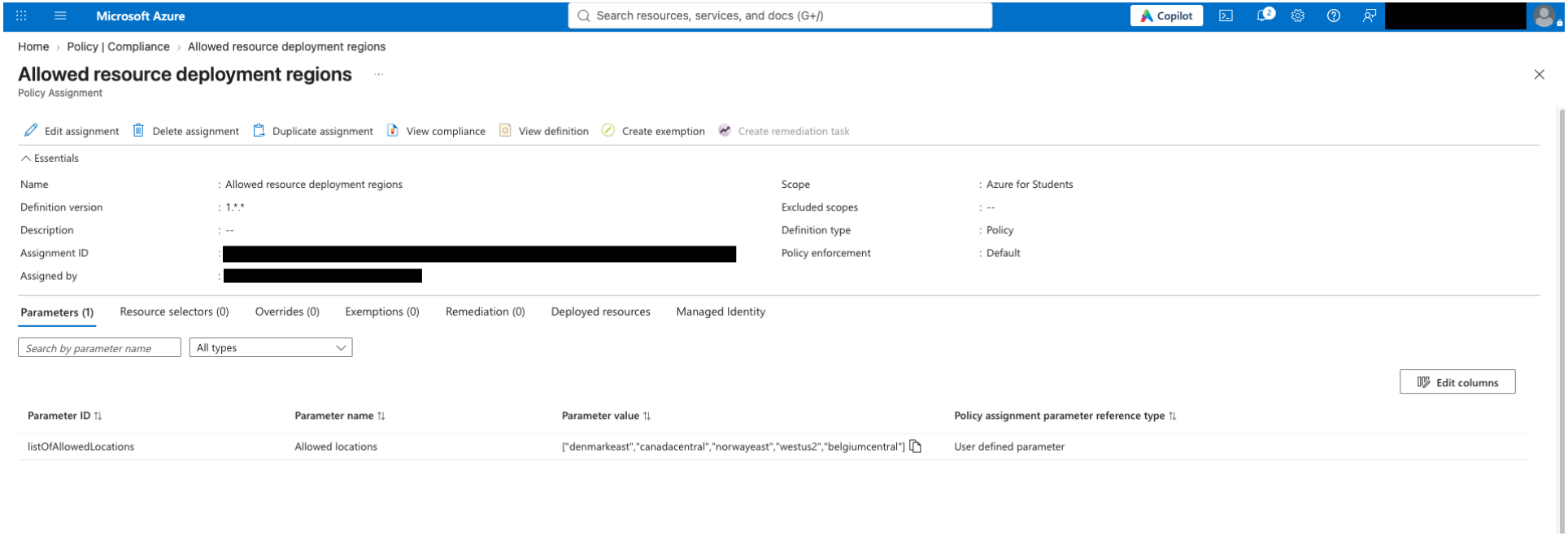
B03b: First VNet deploy denied: East US 2 not allowed by subscription policy. Controls: CM-7, SA-9 (inherited)



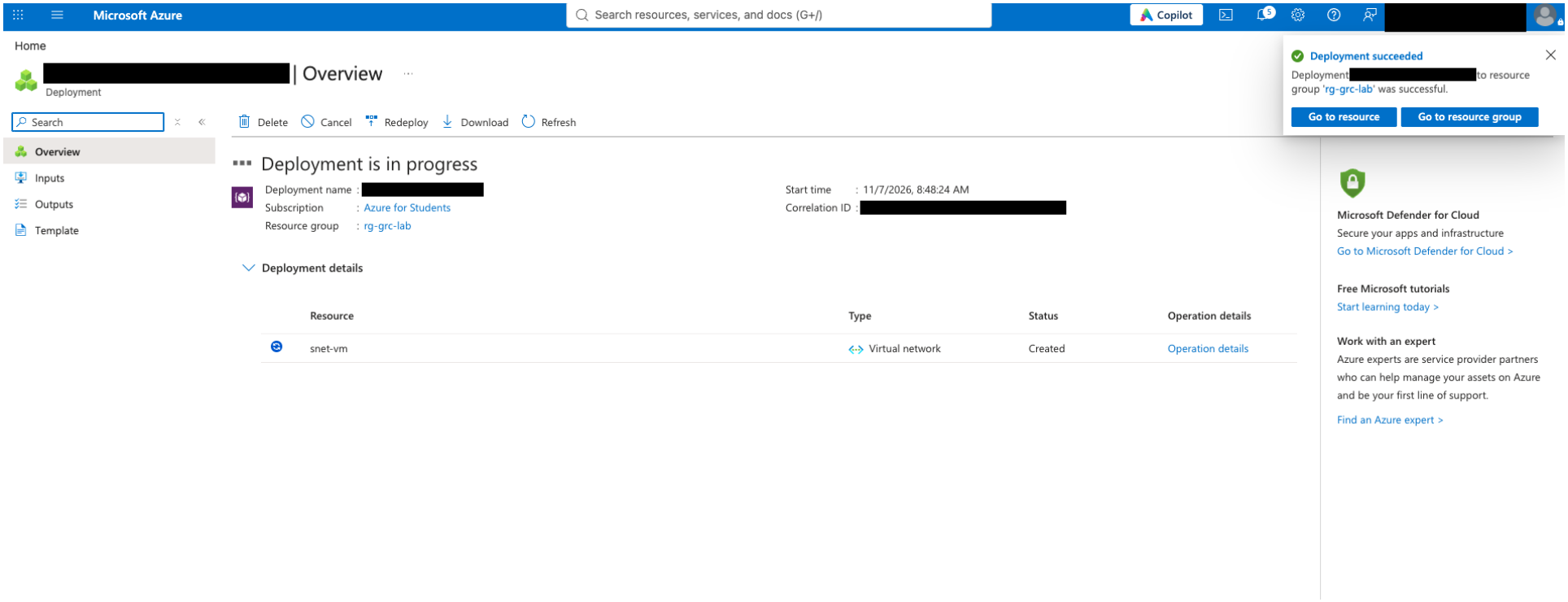
B03c: Root cause: 'Allowed resource deployment regions' policy assignment. Controls: CM-7, SA-9 (inherited)



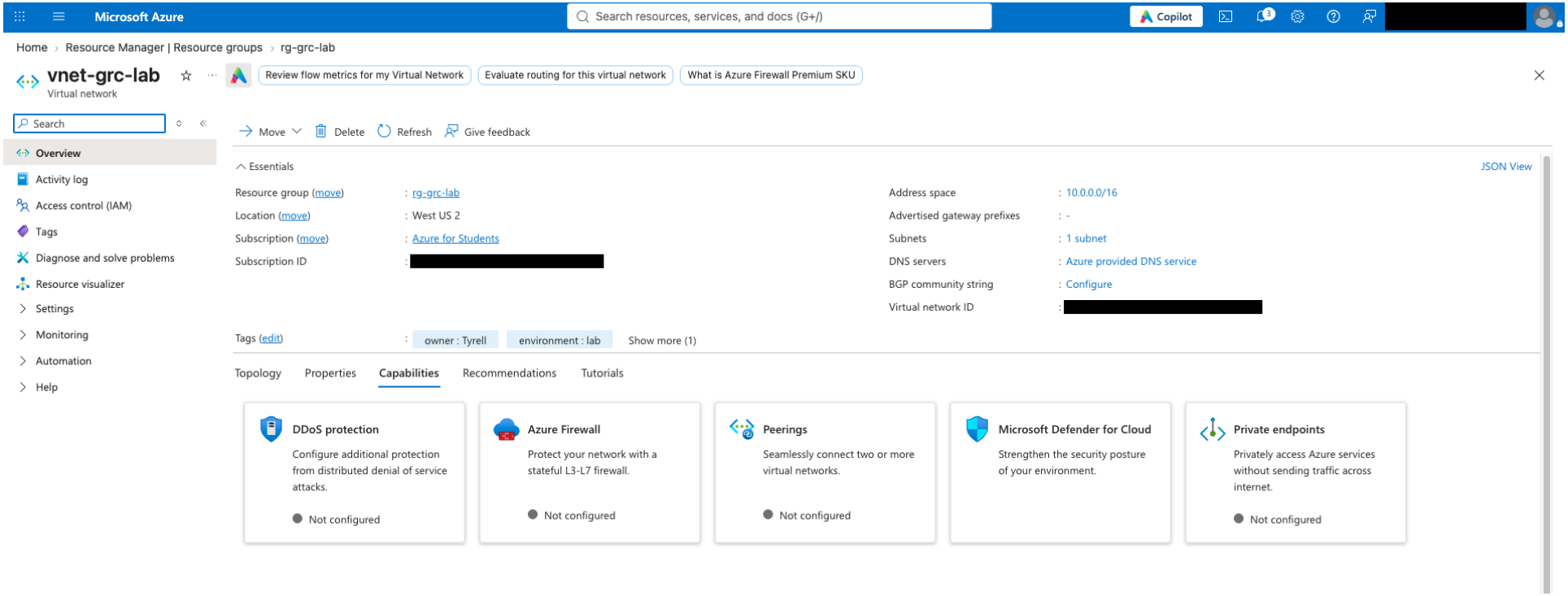
B03d: Allowed regions parameter; West US 2 chosen. Controls: CM-7



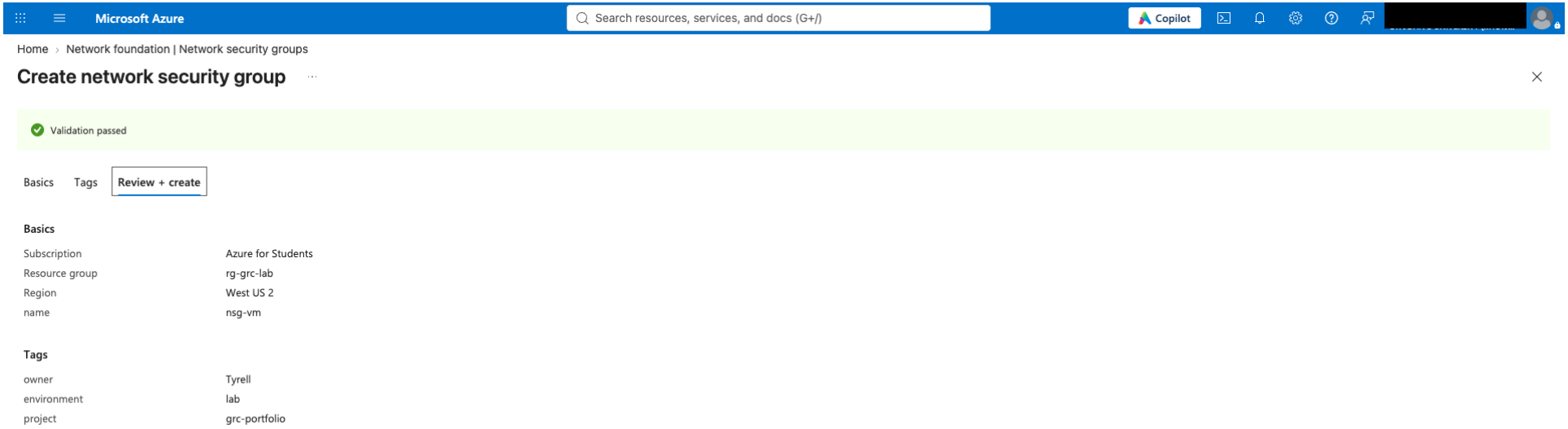
B03e: Mistake caught: VNet created with the wrong name, deleted and rebuilt. Controls: CM-3 (lesson learned)



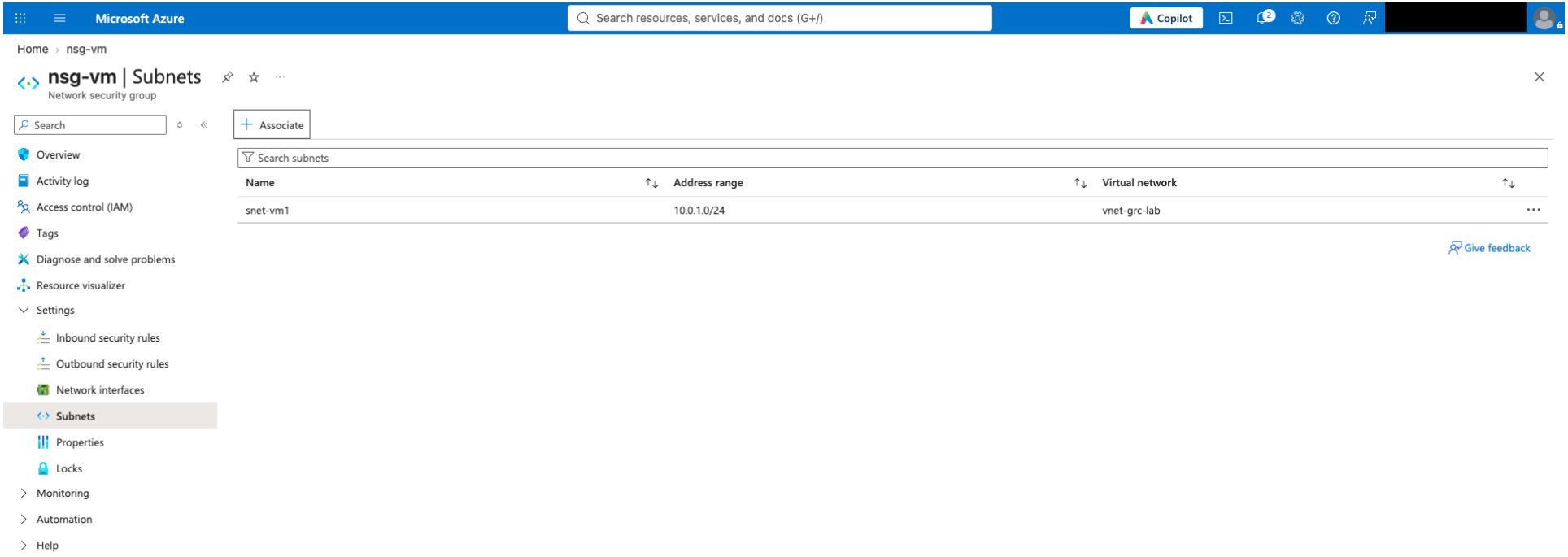
B04: Final VNet vnet-grc-lab, 10.0.0.0/16, West US 2. Controls: SC-7, CM-8



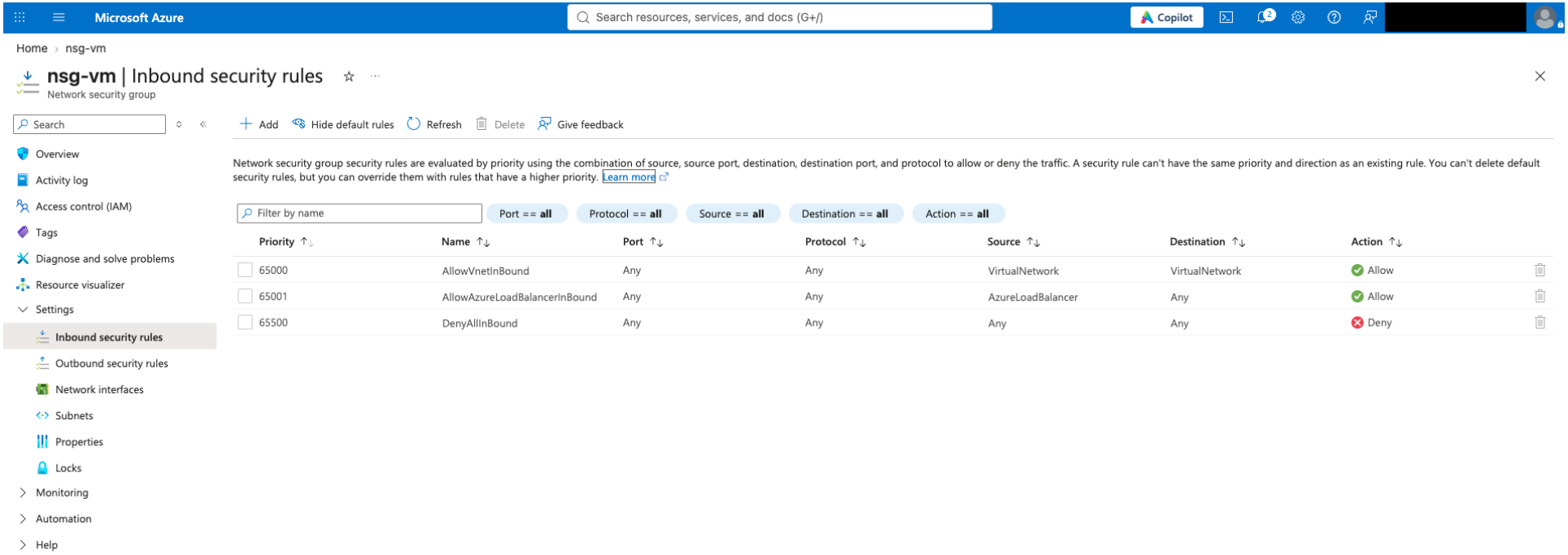
B05: NSG nsg-vm created with tags. Controls: SC-7



B06: nsg-vm associated to subnet snet-vm1. Controls: SC-7

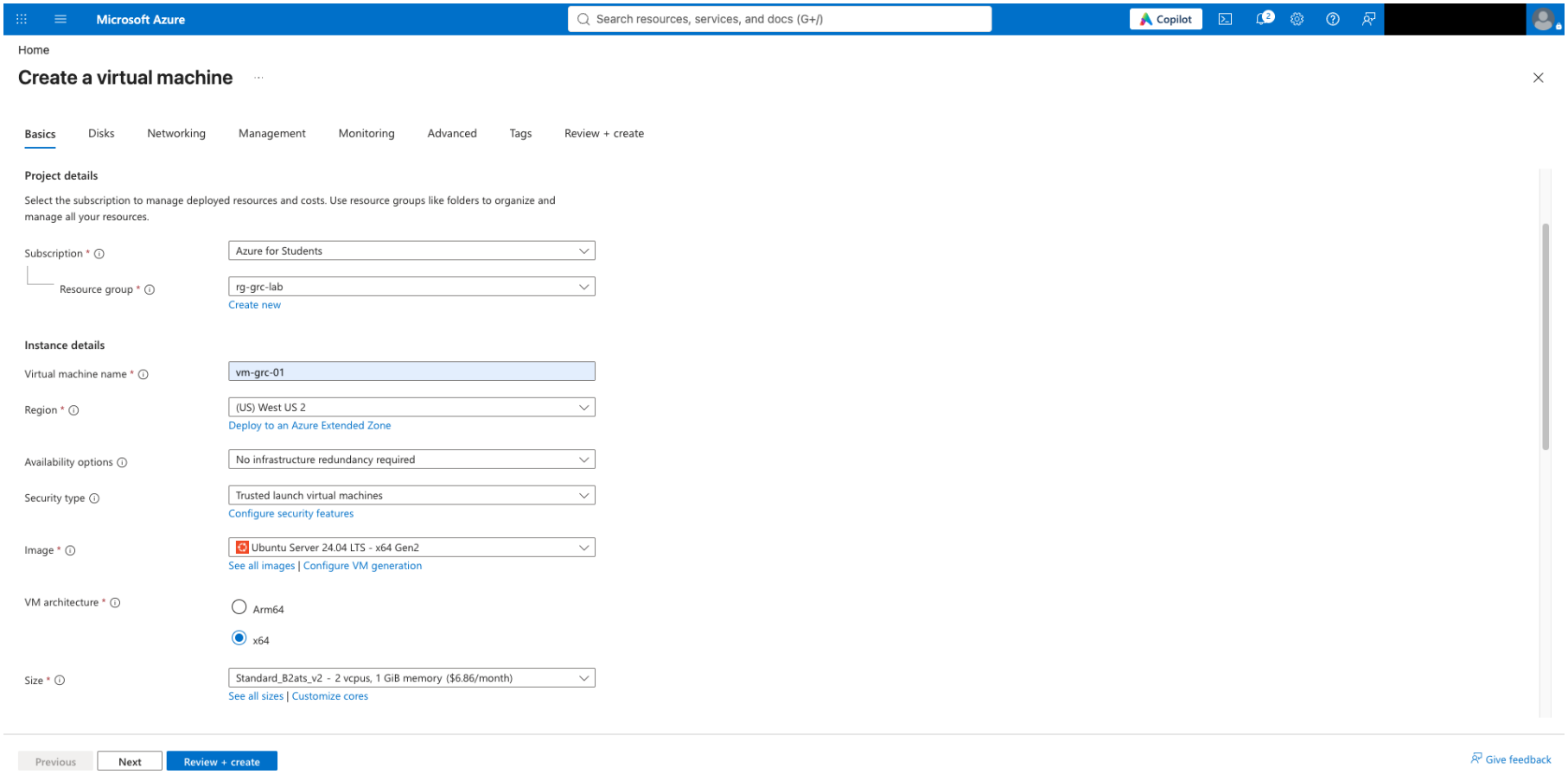


E03: Inbound rules: defaults only, DenyAllInBound at 65500, no custom allow rules. Controls: SC-7, AC-17

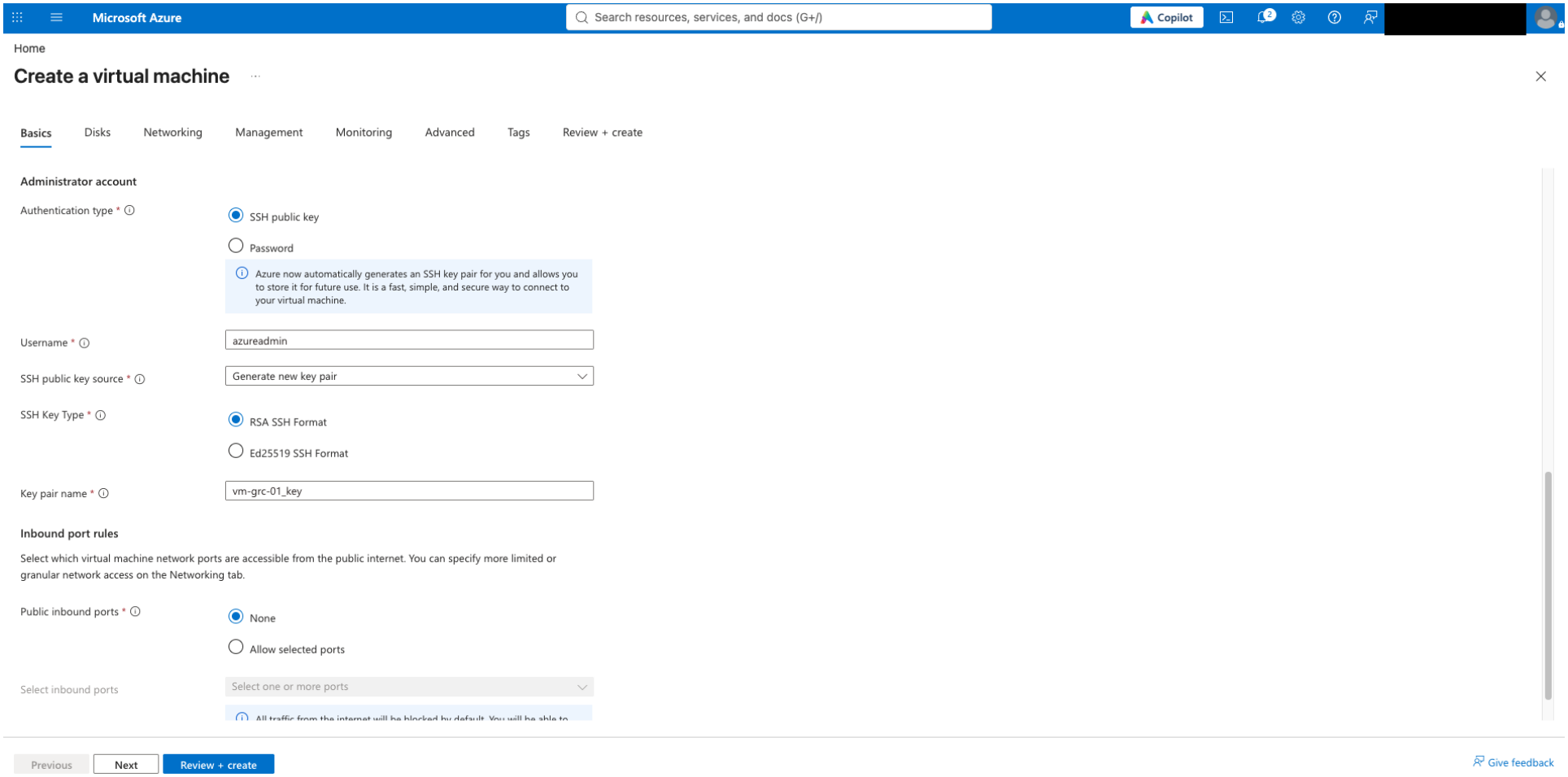


Phase 2. VM build (Nov 7)

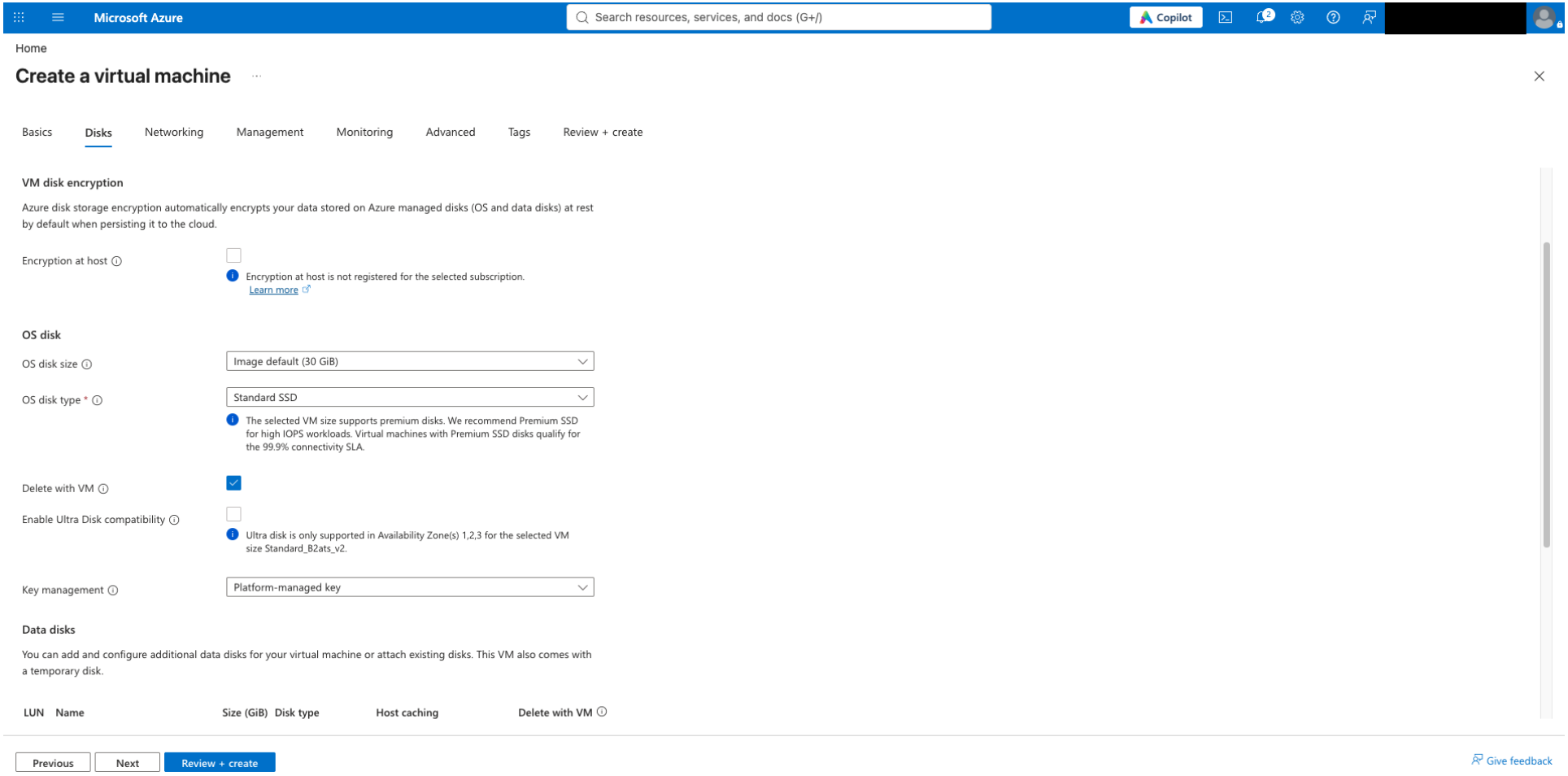
B07a: VM basics: Ubuntu 24.04, Trusted launch, Standard_B2ats_v2, West US 2. Controls: SI-7, CM-2



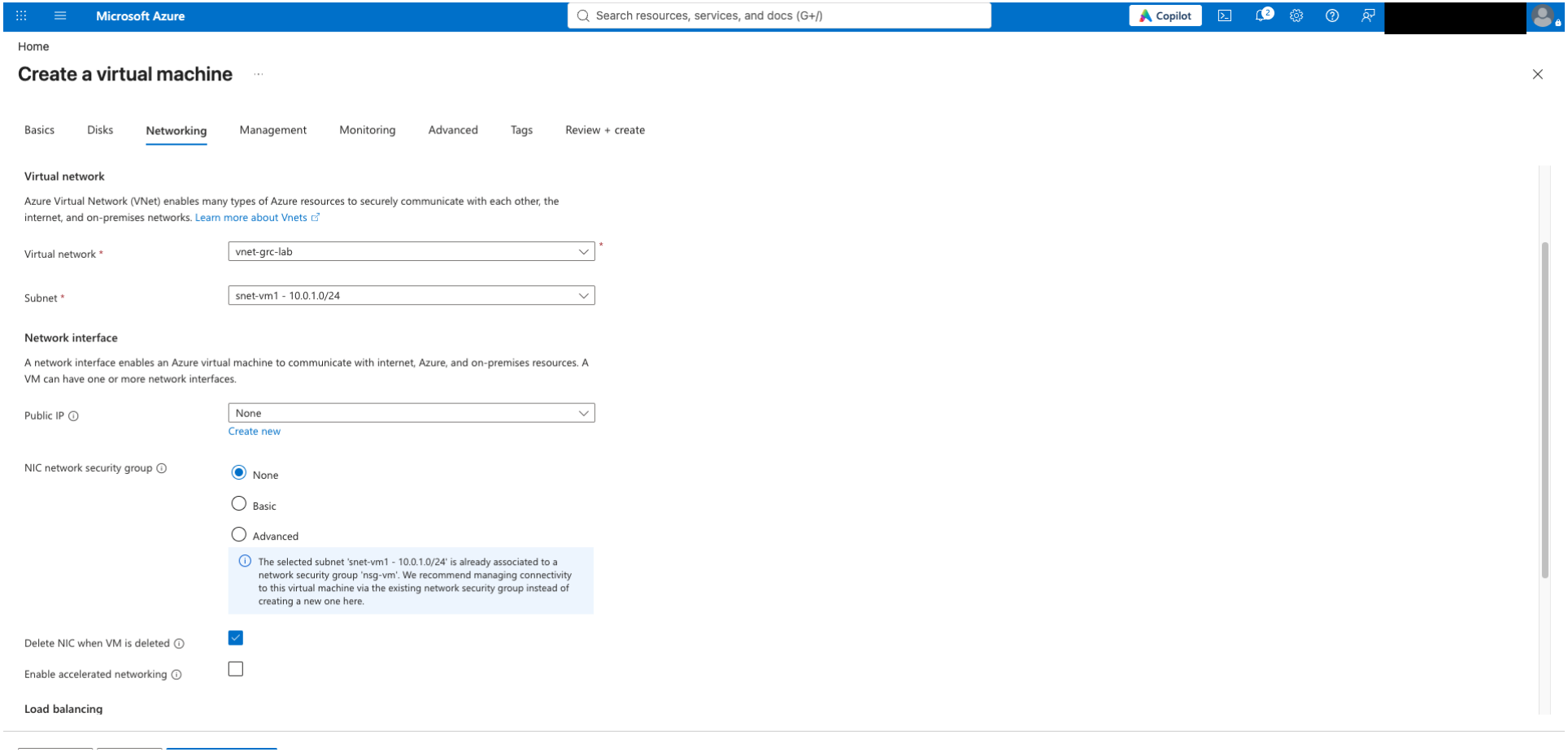
B07b: SSH public key authentication; public inbound ports None. Controls: IA-5, CM-7



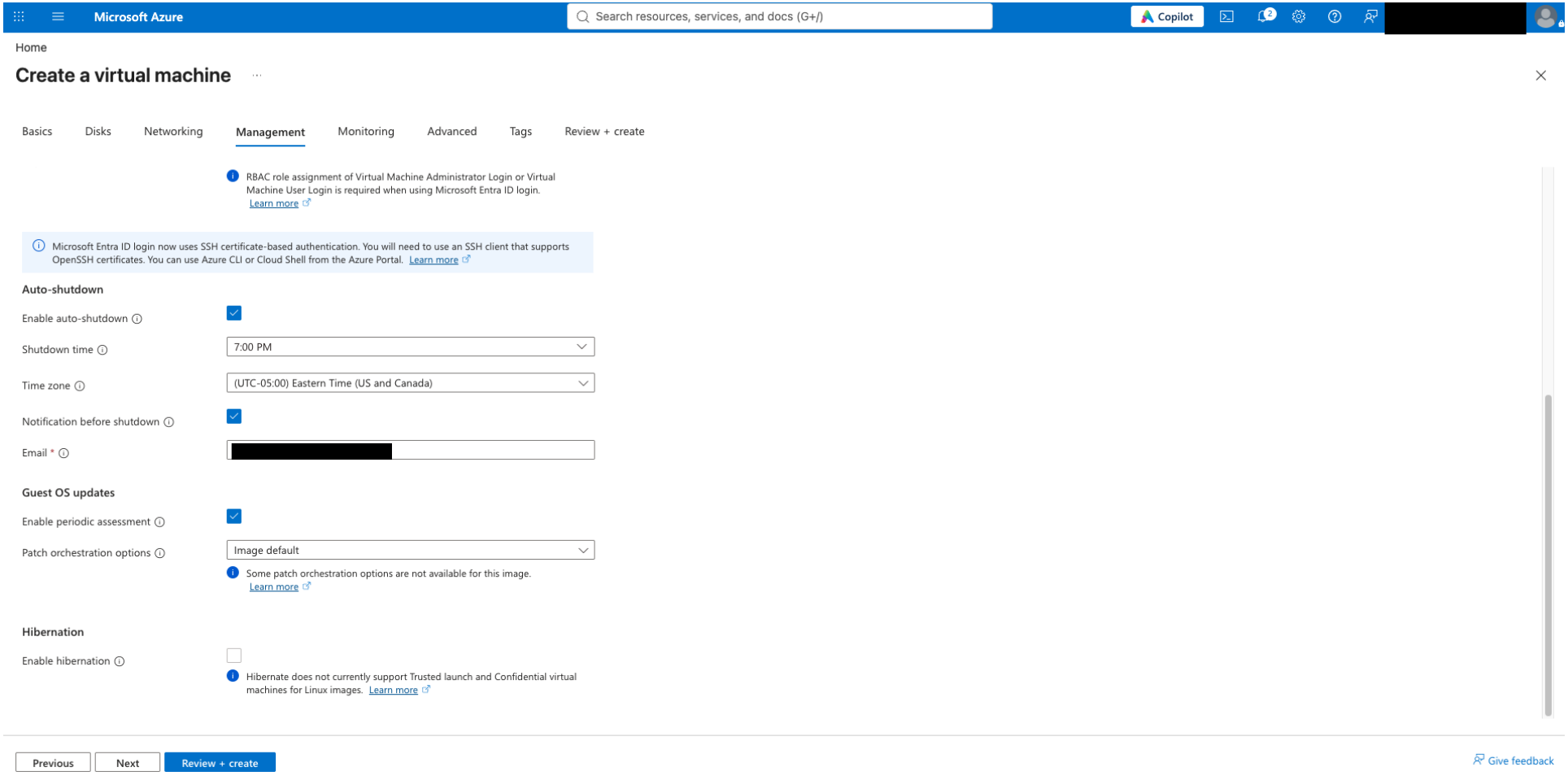
E08: OS disk Standard SSD, encrypted at rest with platform-managed key. Controls: SC-28



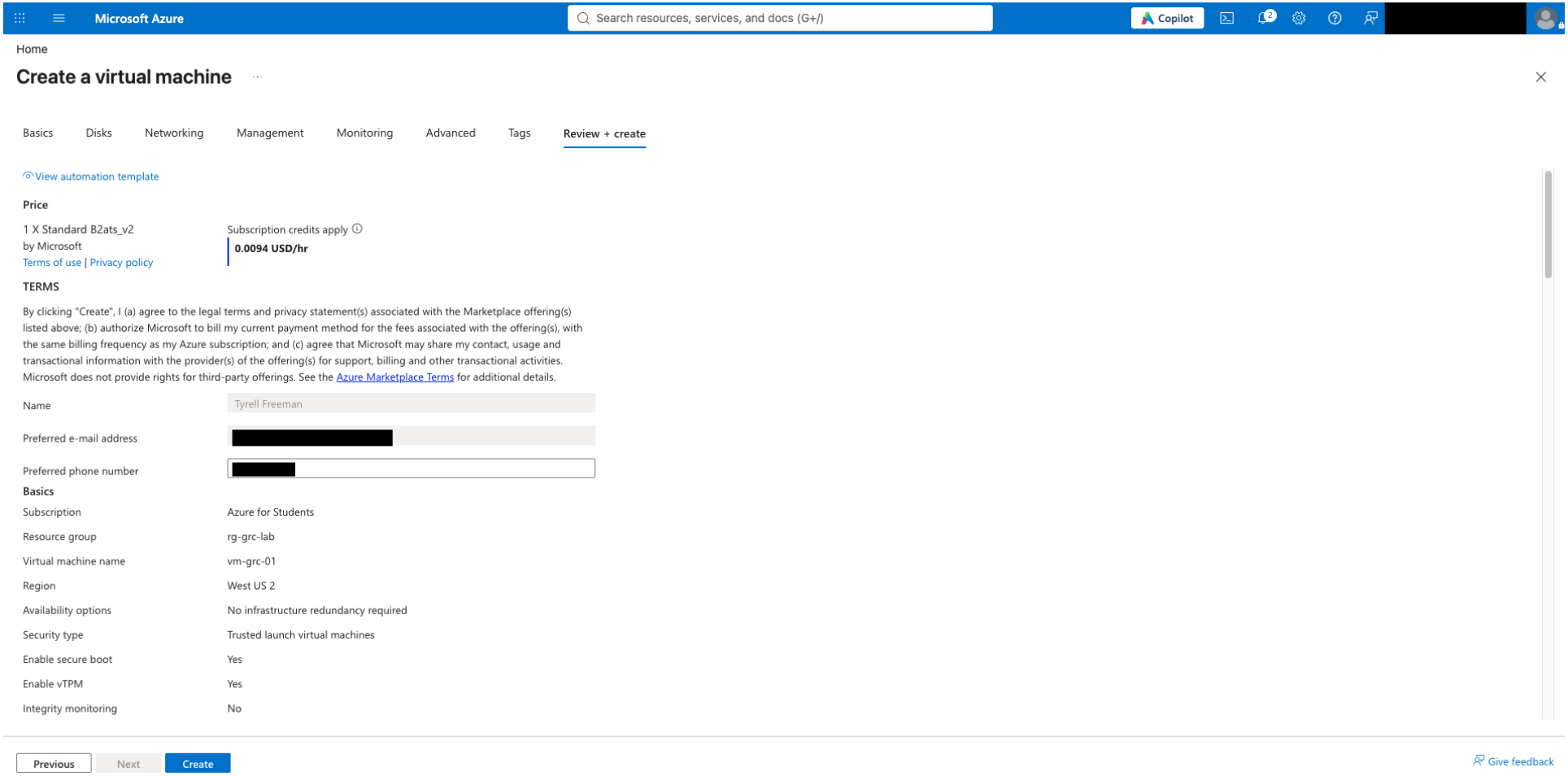
B08: Networking: snet-vm1, Public IP None, NIC NSG None (subnet NSG applies). Controls: SC-7, AC-17



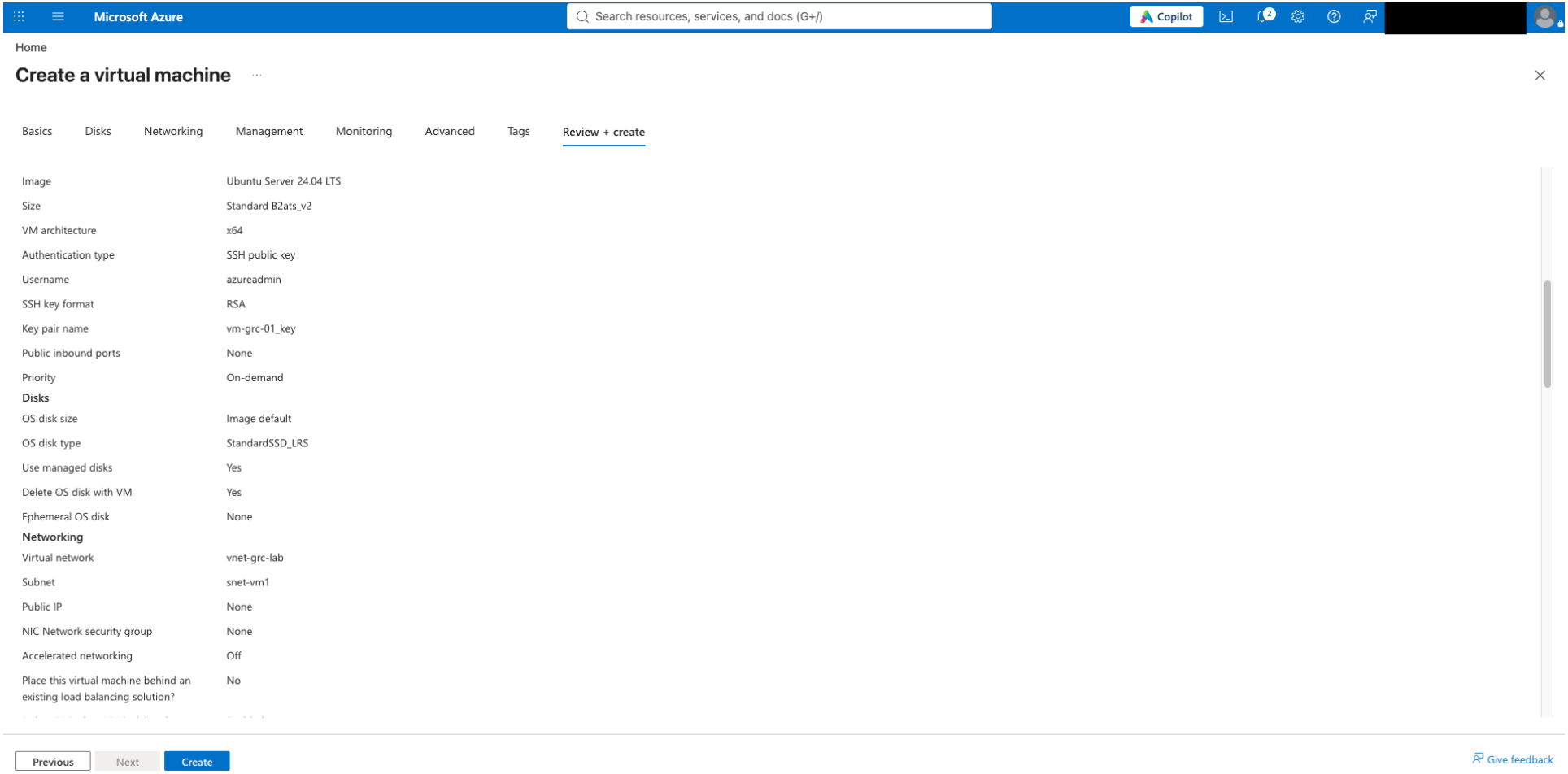
E09: Auto-shutdown 7:00 PM ET; periodic patch assessment on. Controls: SI-2, PM-3



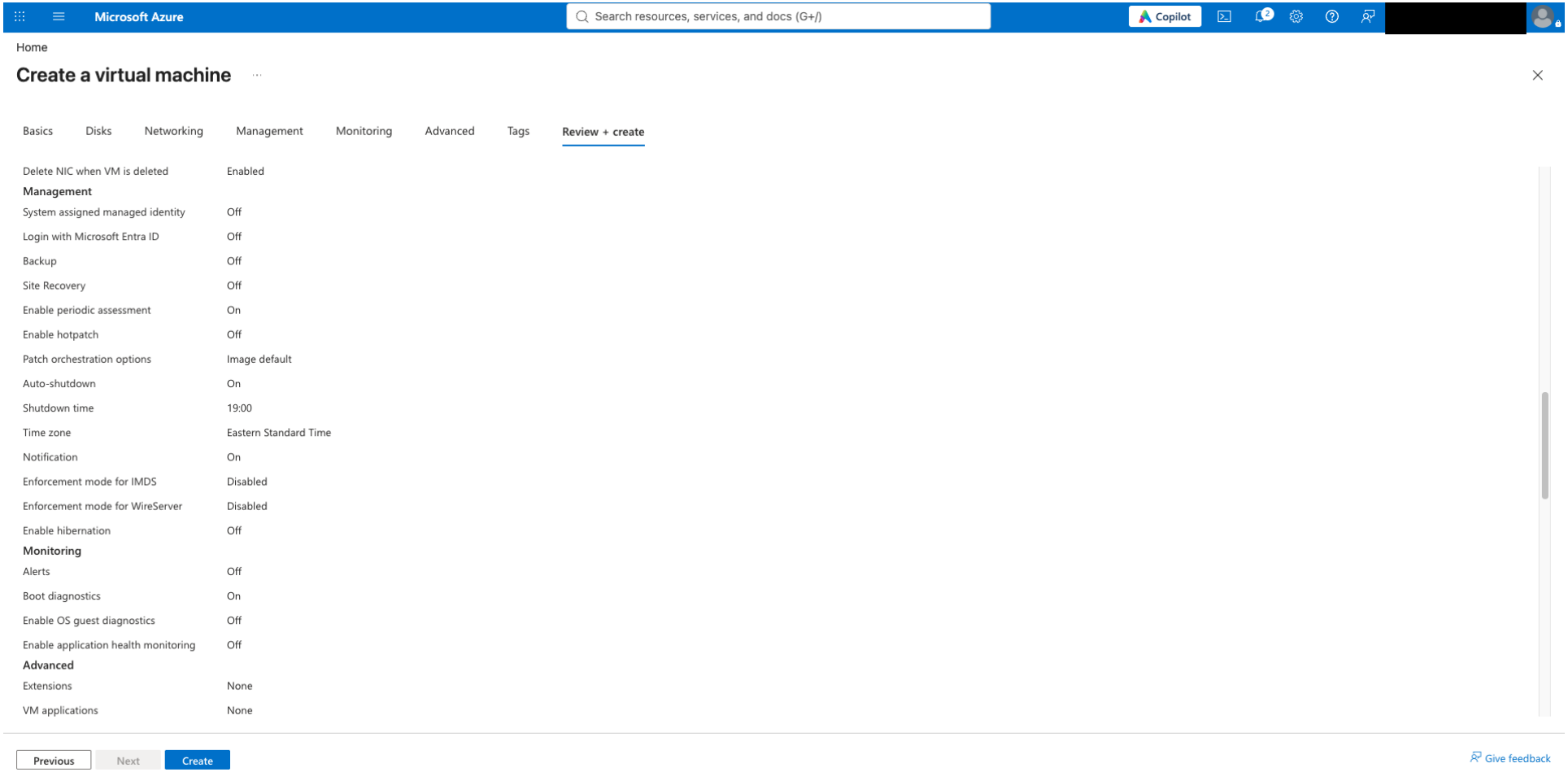
B09a: Final review: price, basics, Trusted launch settings. Controls: CM-2



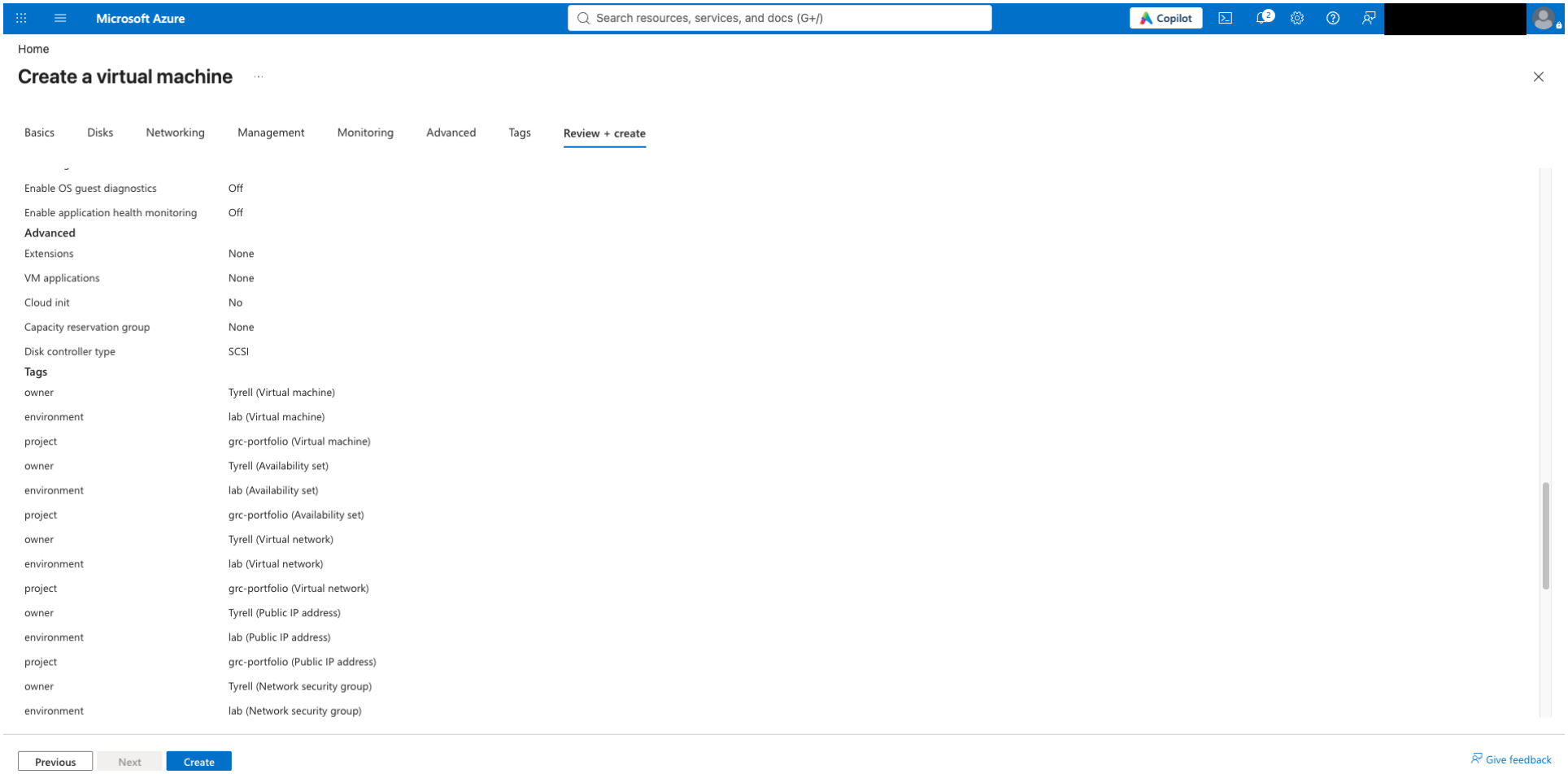
B09b: Final review: authentication, disks, networking. Controls: CM-2



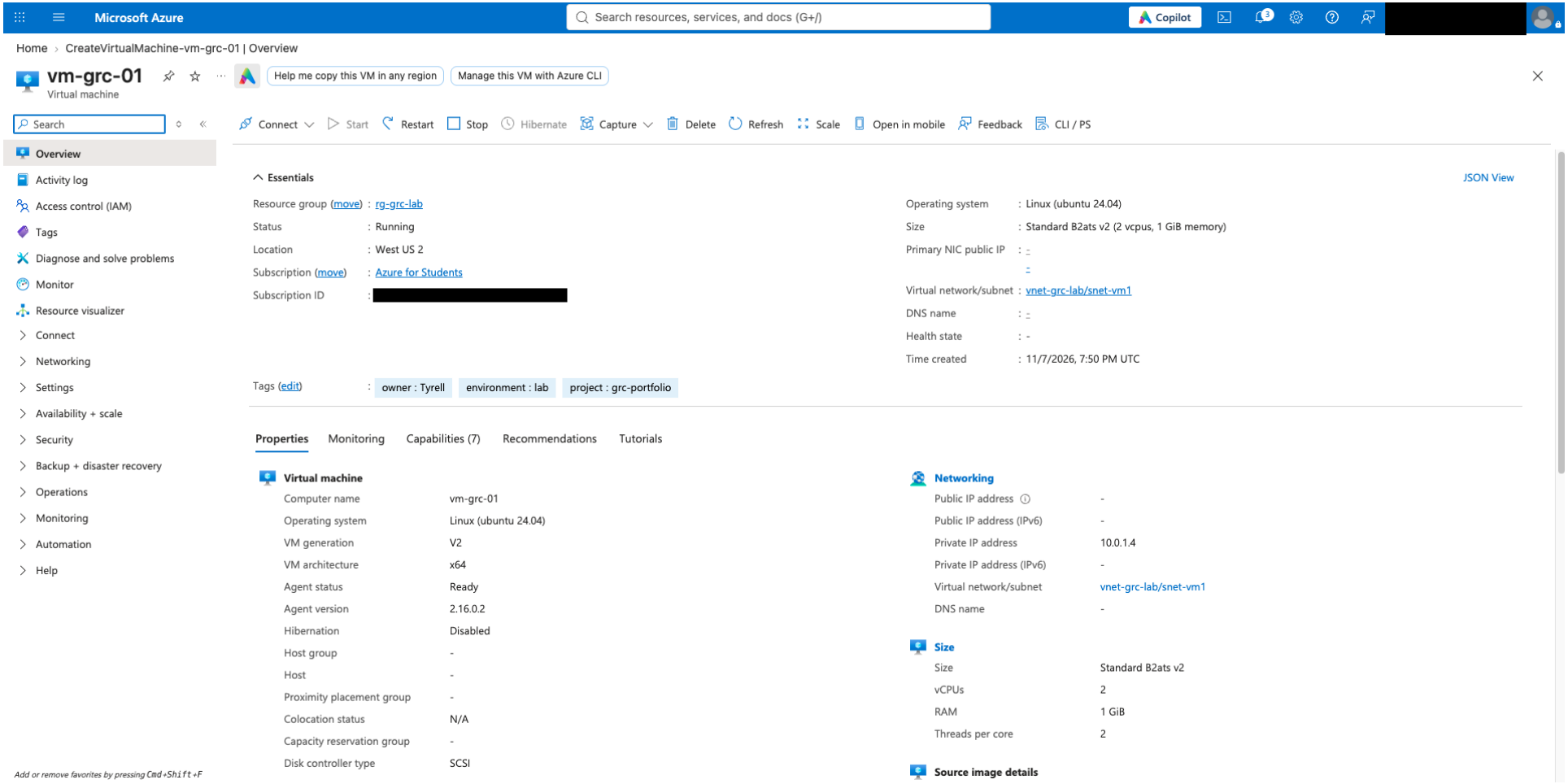
B09c: Final review: management and monitoring settings. Controls: CM-2



B09d: Final review: tags applied to all resources. Controls: CM-8

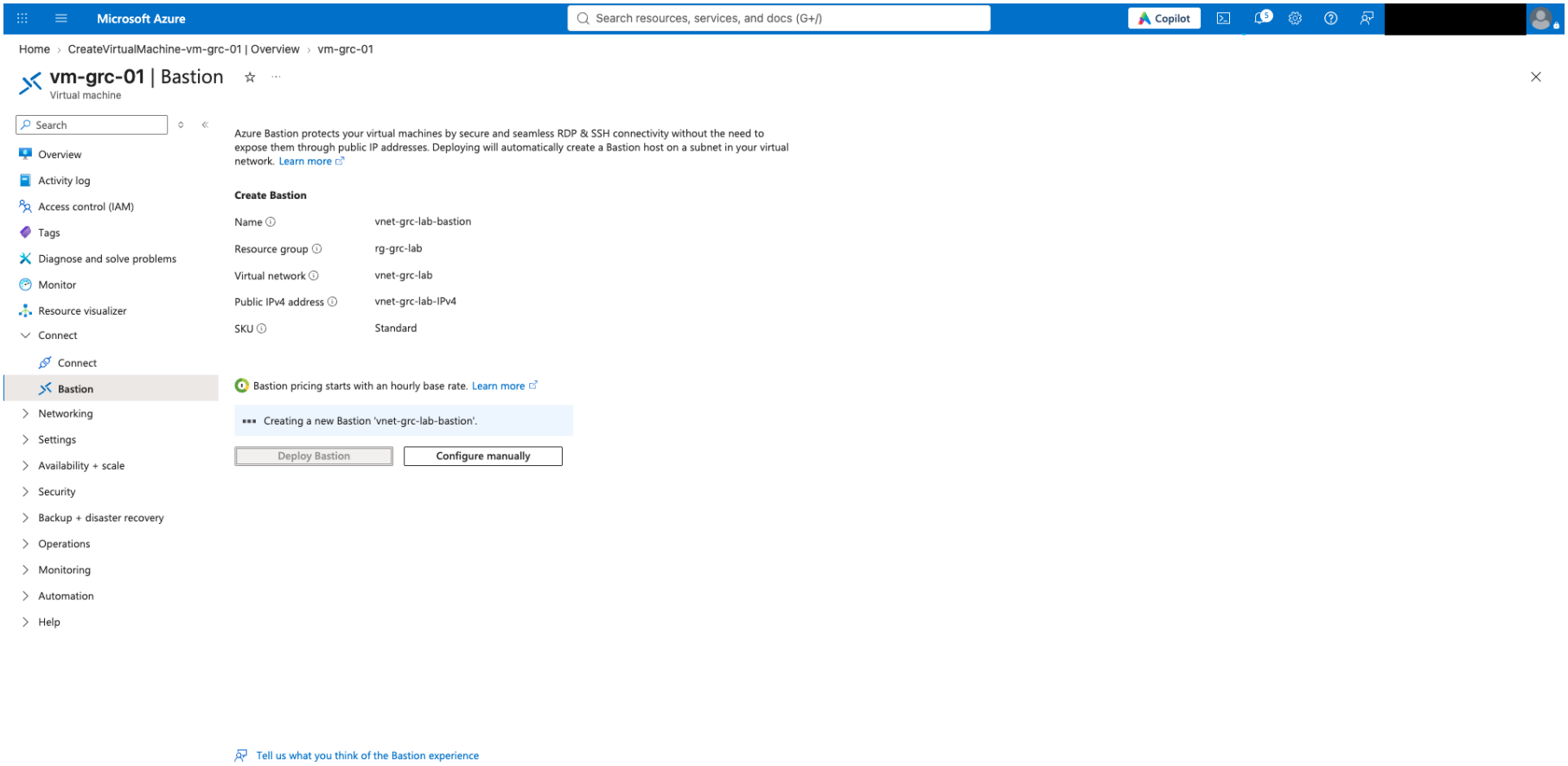


E02: VM running with no public IP; private IP 10.0.1.4 only. Controls: SC-7, AC-17, CM-8

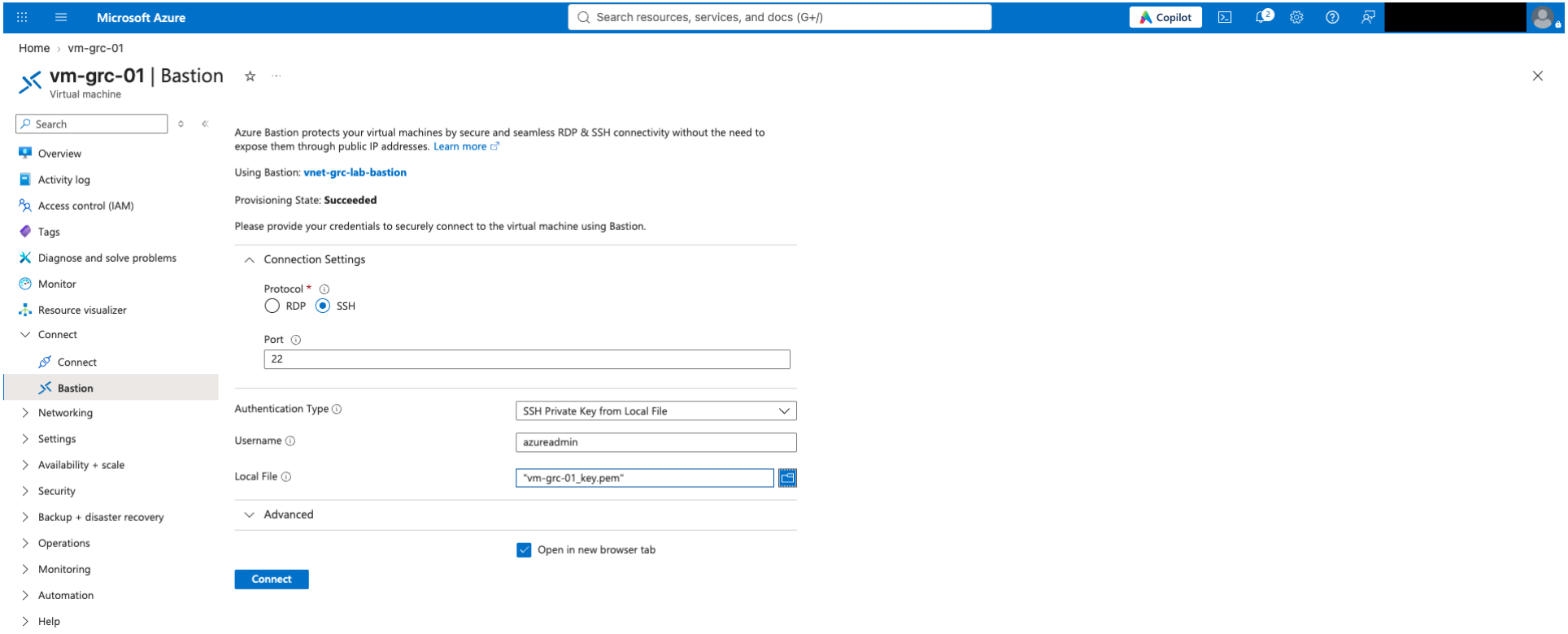


Phase 3. Remote access and patching (Nov 7 to 8)

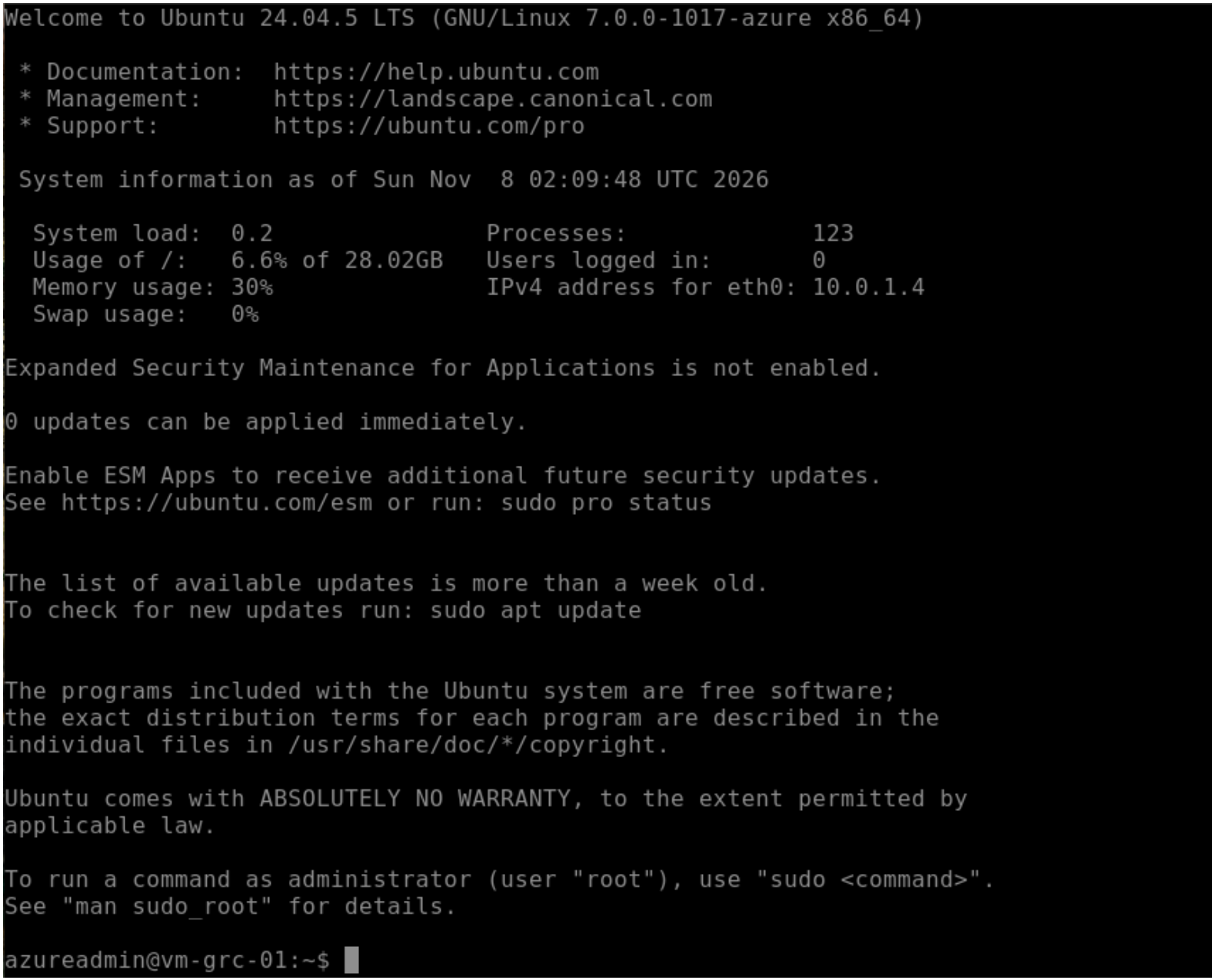
B10: Bastion deployed (Standard SKU; Developer not offered in West US 2). Controls: AC-17(3)



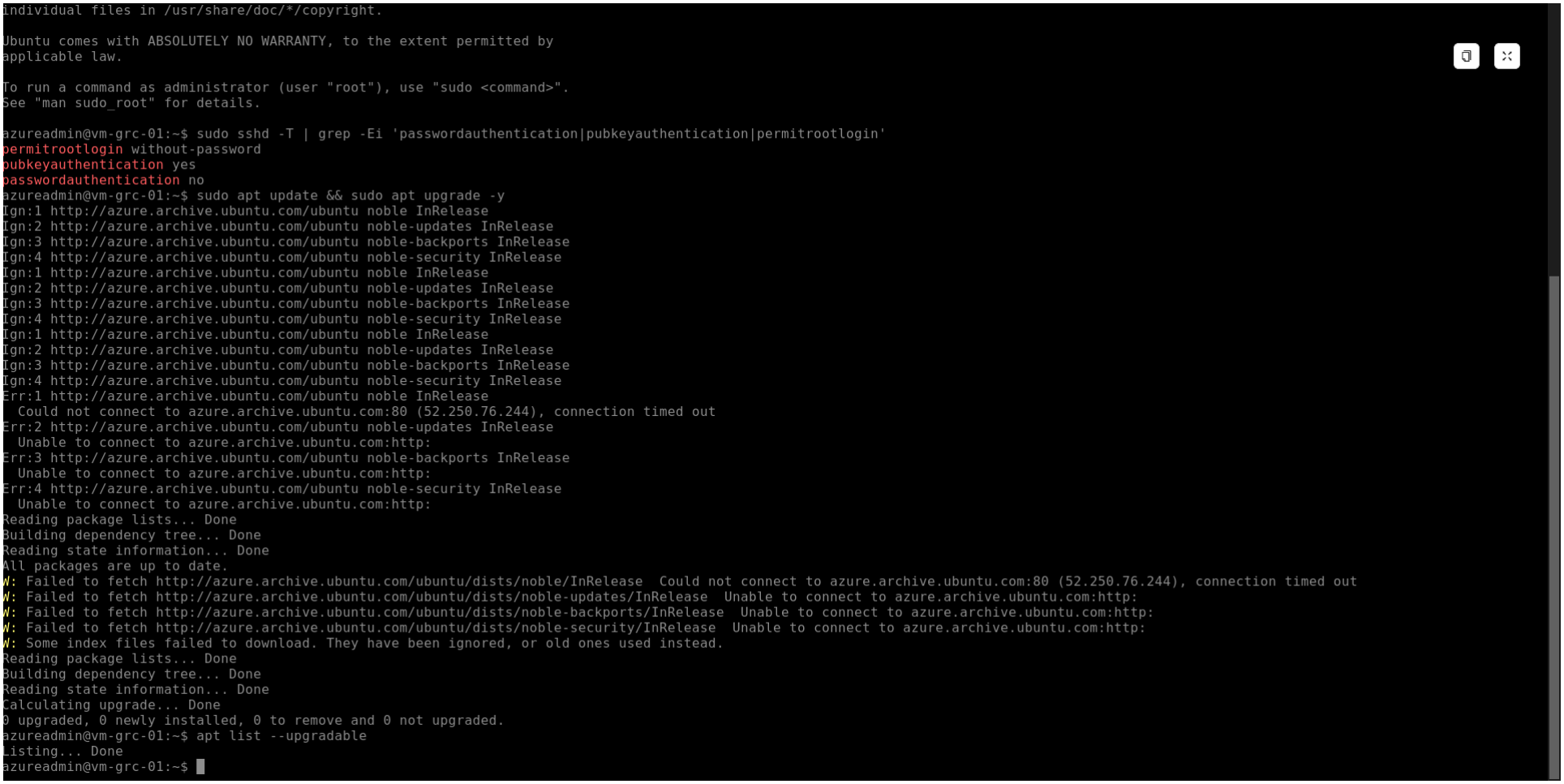
E05a: Bastion connection: SSH port 22, private key from local file, user azureadmin. Controls: AC-17, IA-5



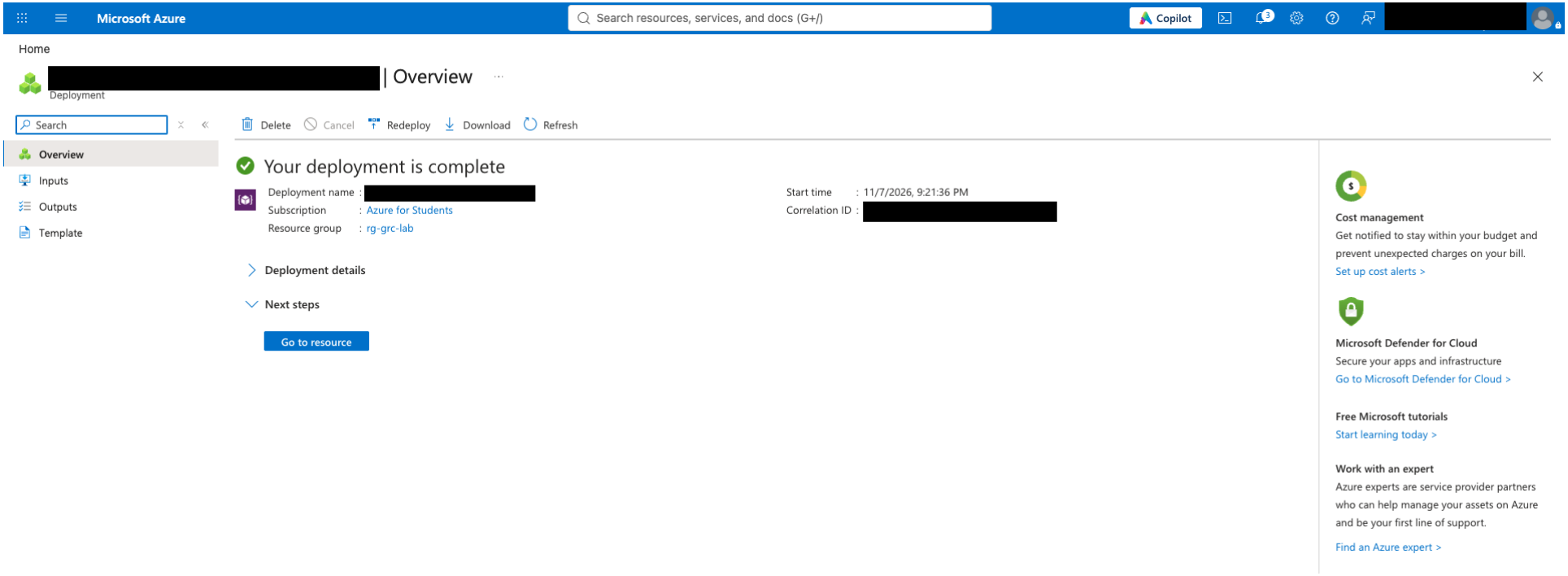
E05b: SSH session established through Bastion to 10.0.1.4. Controls: AC-17(3)



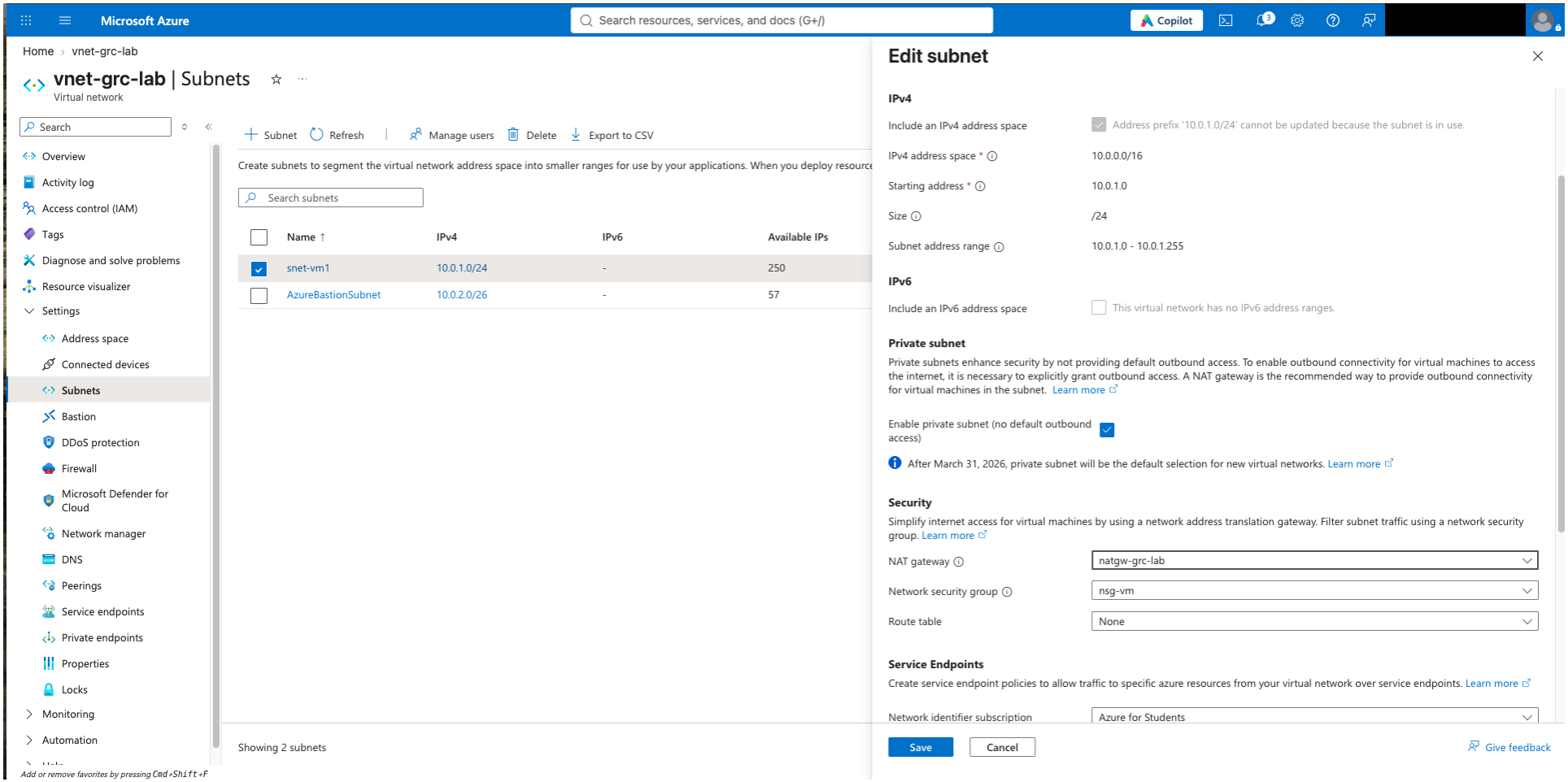
E06 / E07a: sshd: password auth off, key auth on. FINDING: apt said 'up to date' while every repository fetch timed out. Controls: IA-5, SI-2



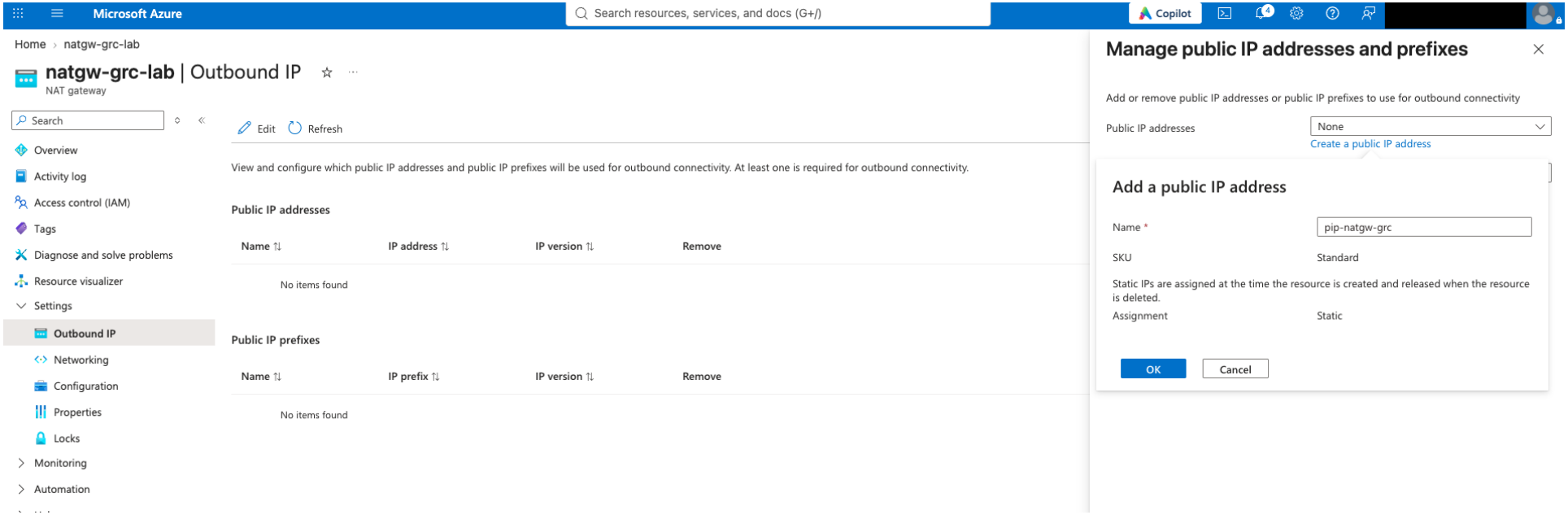
B11: NAT gateway deployed to give the private subnet an egress path. Controls: SC-7



B12: Subnet confirmed private, with NAT gateway and nsg-vm attached. Controls: SC-7



B13: Root cause: NAT gateway had no outbound public IP. Fixed by adding one. Controls: SI-2 (root cause)



E07b: apt upgrade succeeds after the fix; updates installed. Controls: SI-2

```
Unpacking libmspack0t64:amd64 (0.11-1.1ubuntu0.24.04.1) over (0.11-1.1build1) ...
Preparing to unpack .../7-linux-cloud-tools-common_6.8.0-149.149_all.deb ...
Unpacking linux-cloud-tools-common (6.8.0-149.149) over (6.8.0-147.147) ...
Preparing to unpack .../8-linux-tools-common_6.8.0-149.149_all.deb ...
Unpacking linux-tools-common (6.8.0-149.149) over (6.8.0-147.147) ...
Preparing to unpack .../9-python3-twisted_24.3.0-1ubuntu0.3_all.deb ...
Unpacking python3-twisted (24.3.0-1ubuntu0.3) over (24.3.0-1ubuntu0.2) ...
Setting up apt-utils (2.8.4ubuntu0.1) ...
Setting up bind9-libs:amd64 (1:9.18.39-0ubuntu0.24.04.9) ...
Setting up libbrotli1:amd64 (1.1.0-2ubuntu0.1) ...
Setting up libsqlite3-0:amd64 (3.45.1-1ubuntu2.9) ...
Setting up libmspack0t64:amd64 (0.11-1.1ubuntu0.24.04.1) ...
Setting up linux-cloud-tools-common (6.8.0-149.149) ...
Setting up ca-certificates (20260816ubuntu0.24.04.2) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Setting up python3-twisted (24.3.0-1ubuntu0.3) ...
Setting up bind9-host (1:9.18.39-0ubuntu0.24.04.9) ...
Setting up linux-tools-common (6.8.0-149.149) ...
Setting up bind9-dnsutils (1:9.18.39-0ubuntu0.24.04.9) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
 /etc/needrestart/restart.d/systemd-manager
 systemctl restart cron.service fwupd.service multipathd.service packagekit.service polkit.service ssh.service systemd-journald.service systemd-networkd.service systemd-resolved.service systemd-udevd.service udisks2.service

Service restarts being deferred:
 systemctl restart ModemManager.service
 /etc/needrestart/restart.d/dbus.service
 systemctl restart networkd-dispatcher.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 azureadmin @ session #2: apt[2250], sshd[1063]
 azureadmin @ user manager service: systemd[1068]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureadmin@vm-grc-01:~$
```

E07c: apt list --upgradable returns nothing: fully patched. Controls: SI-2

```
Setting up ca-certificates (20260816ubuntu0.24.04.2) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Setting up python3-twisted (24.3.0-1ubuntu0.3) ...
Setting up bind9-host (1:9.18.39-0ubuntu0.24.04.9) ...
Setting up linux-tools-common (6.8.0-149.149) ...
Setting up bind9-dnsutils (1:9.18.39-0ubuntu0.24.04.9) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
 /etc/needrestart/restart.d/systemd-manager
 systemctl restart cron.service fwupd.service multipathd.service packagekit.service polkit.service ssh.service systemd-journald.service systemd-networkd.service systemd-resolved.service systemd-udevd.service udisks2.service

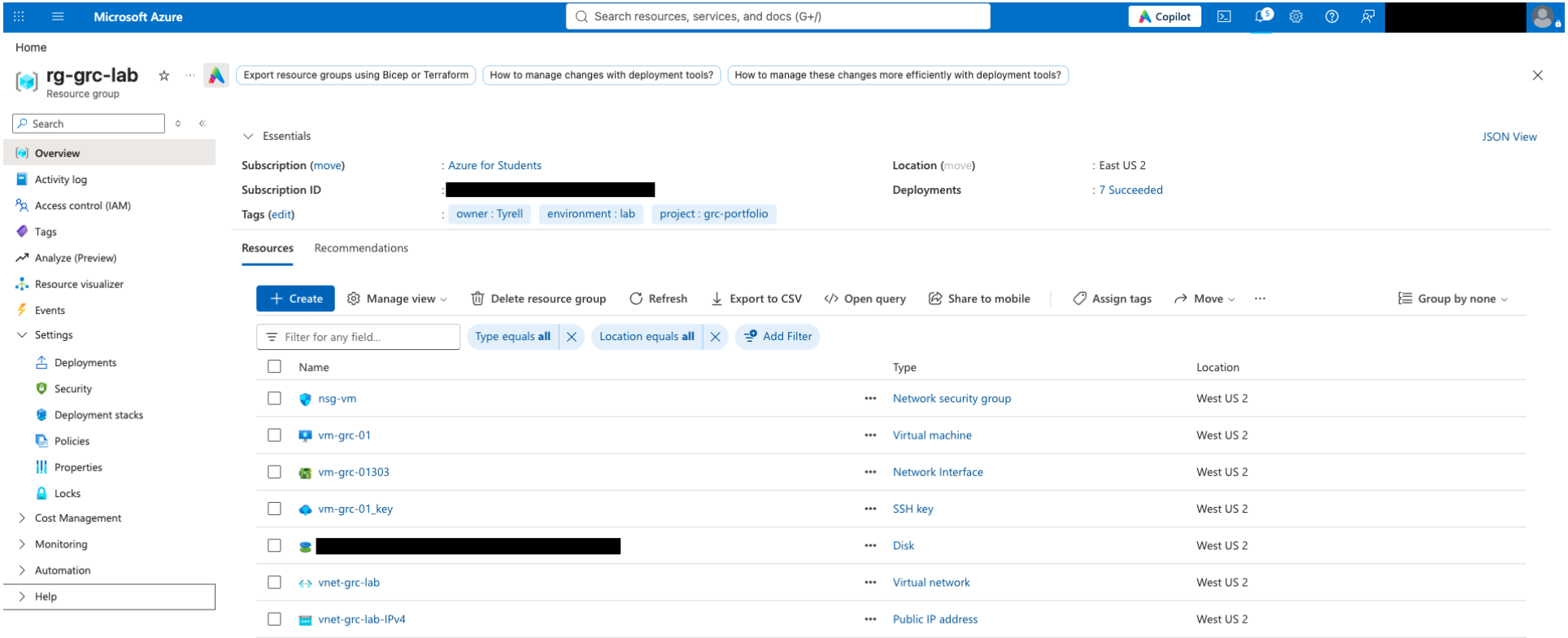
Service restarts being deferred:
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 systemctl restart networkd-dispatcher.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

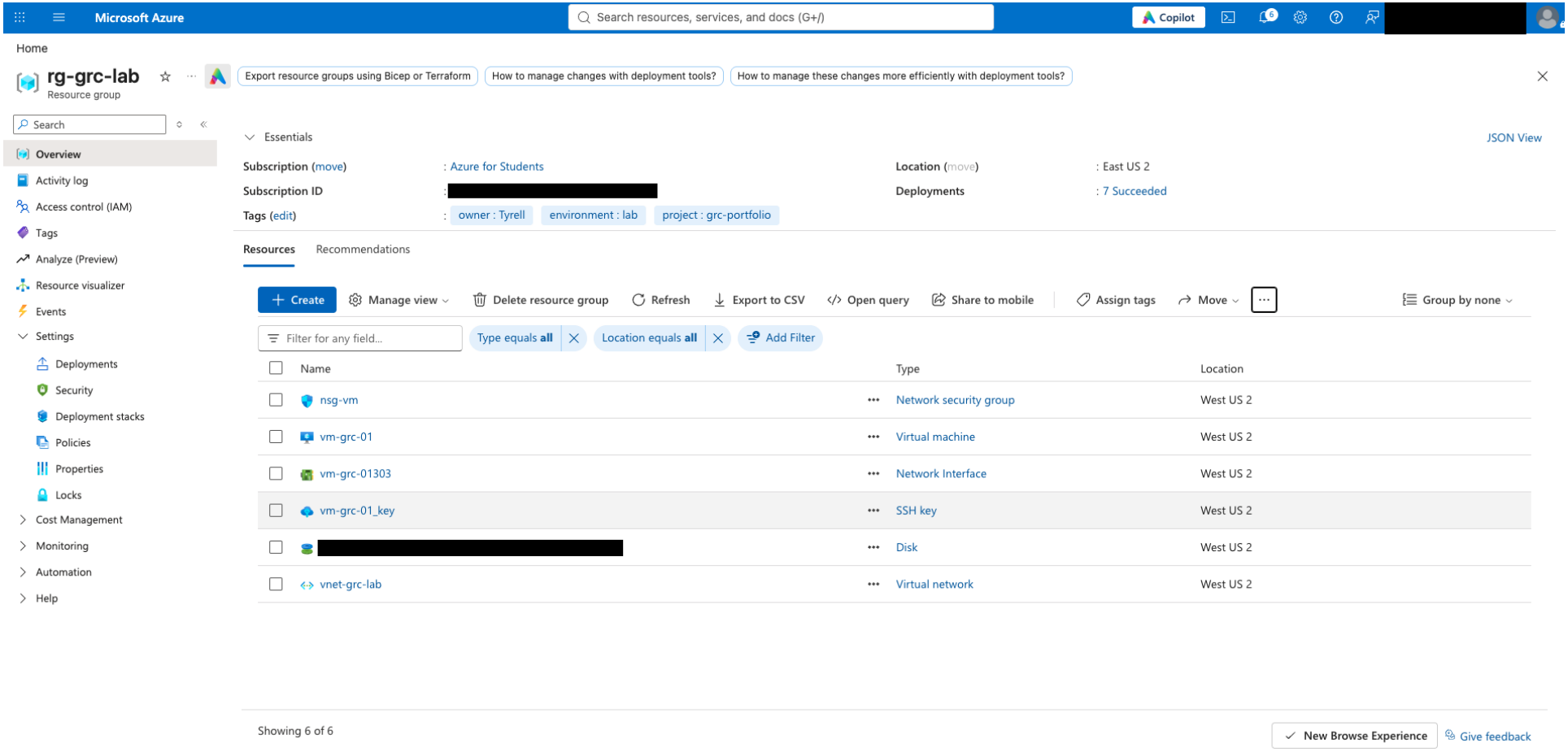
User sessions running outdated binaries:
 azureadmin @ session #2: apt[2250], sshd[1063]
 azureadmin @ user manager service: systemd[1068]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureadmin@vm-grc-01:~$ apt list --upgradable
Listing... Done
azureadmin@vm-grc-01:~$
```

B14: Teardown in progress: Bastion and NAT gateway removed. Controls: PM-3

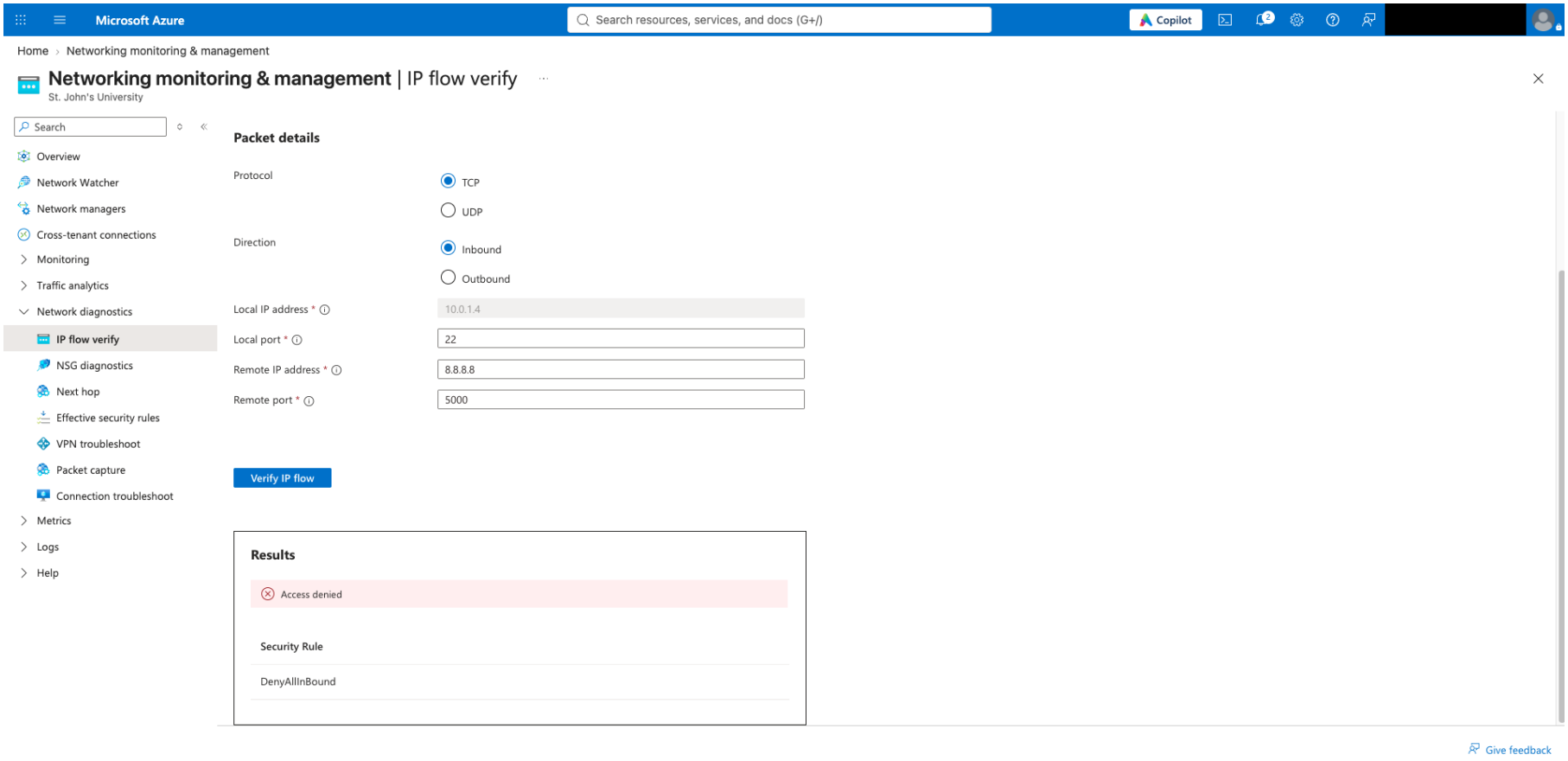


B14b: Teardown complete: 6 resources, no public IPs. Controls: PM-3, CM-8

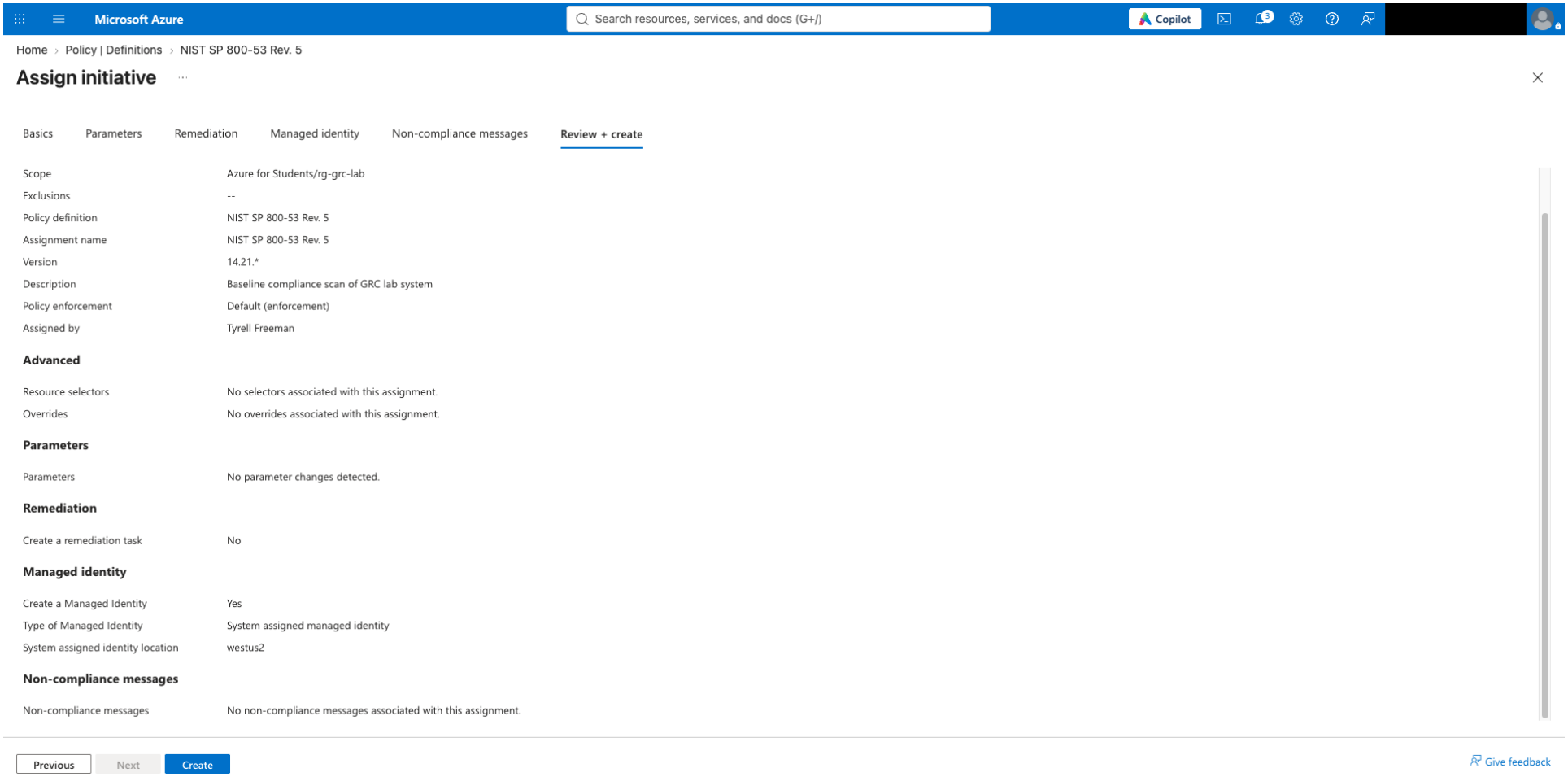


Phase 4. Assessment (Nov 9)

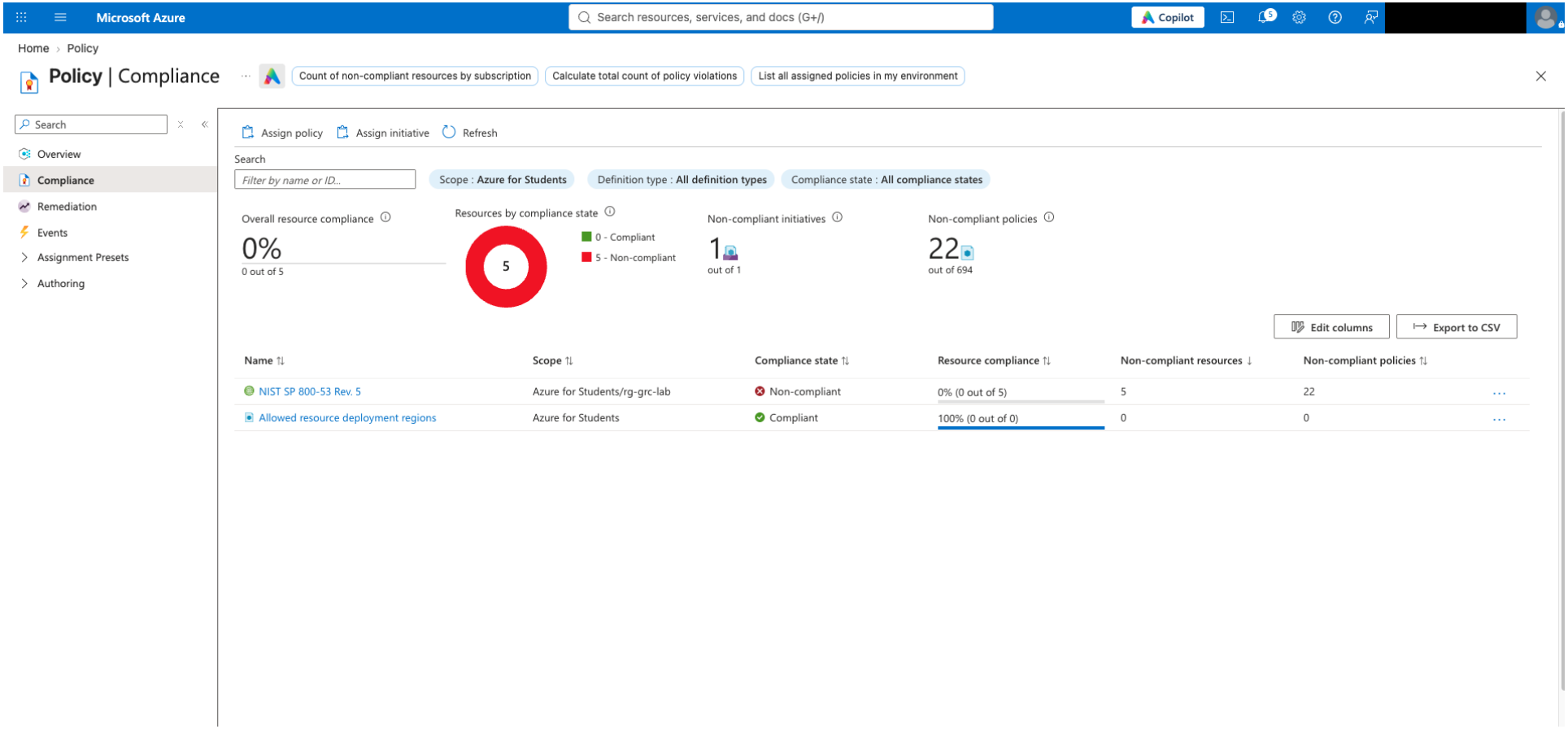
E04: IP flow verify: internet to TCP 22 = Access denied by DenyAllInBound. Controls: SC-7, AC-17, CA-7



E11a: NIST SP 800-53 Rev. 5 initiative assigned to rg-grc-lab. Controls: CA-2, CA-7

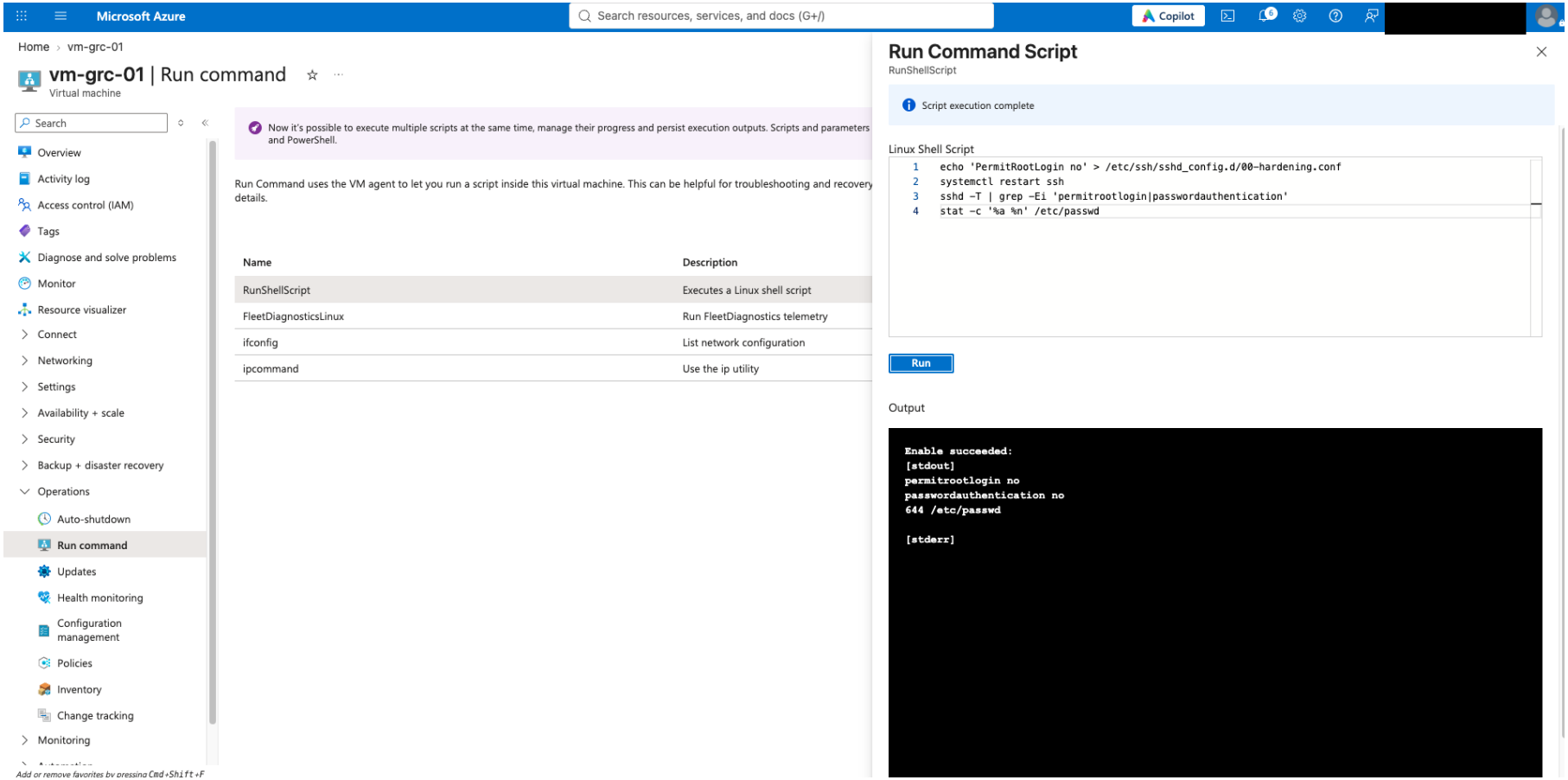


E11b: Baseline scan: 22 of 694 policies non-compliant (triaged in the POA&M; workbook). Controls: CA-2, CA-7, RA-5

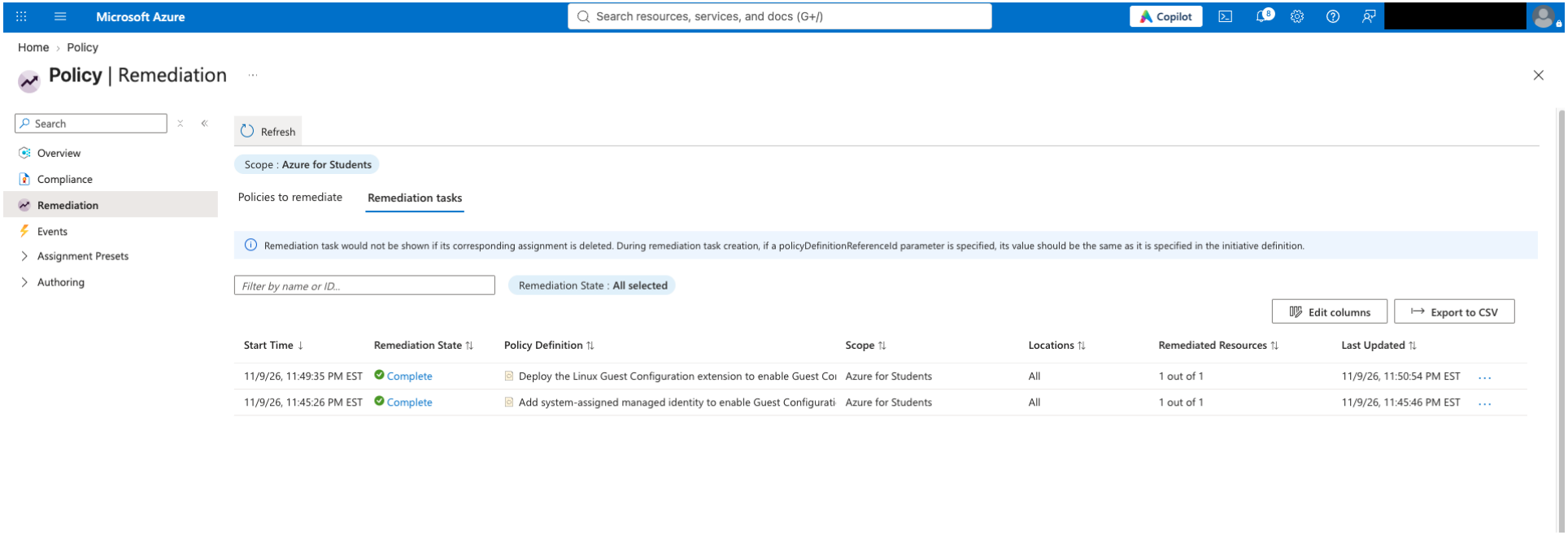


Phase 5. POA&M; remediation (Nov 9 to 10)

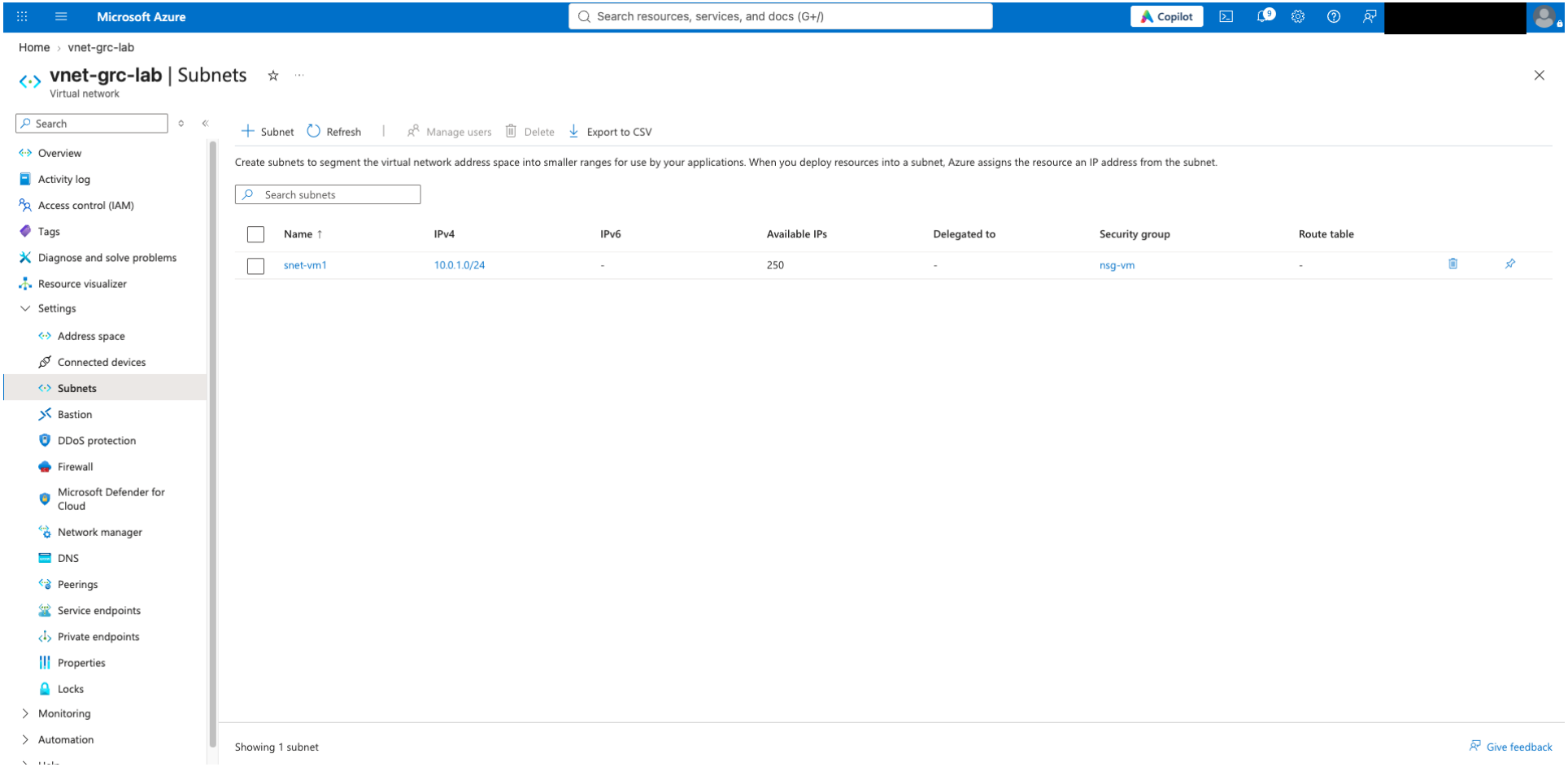
F07: POAM-07 closed: root login disabled; /etc/passwd 644. Controls: CM-6, AC-6



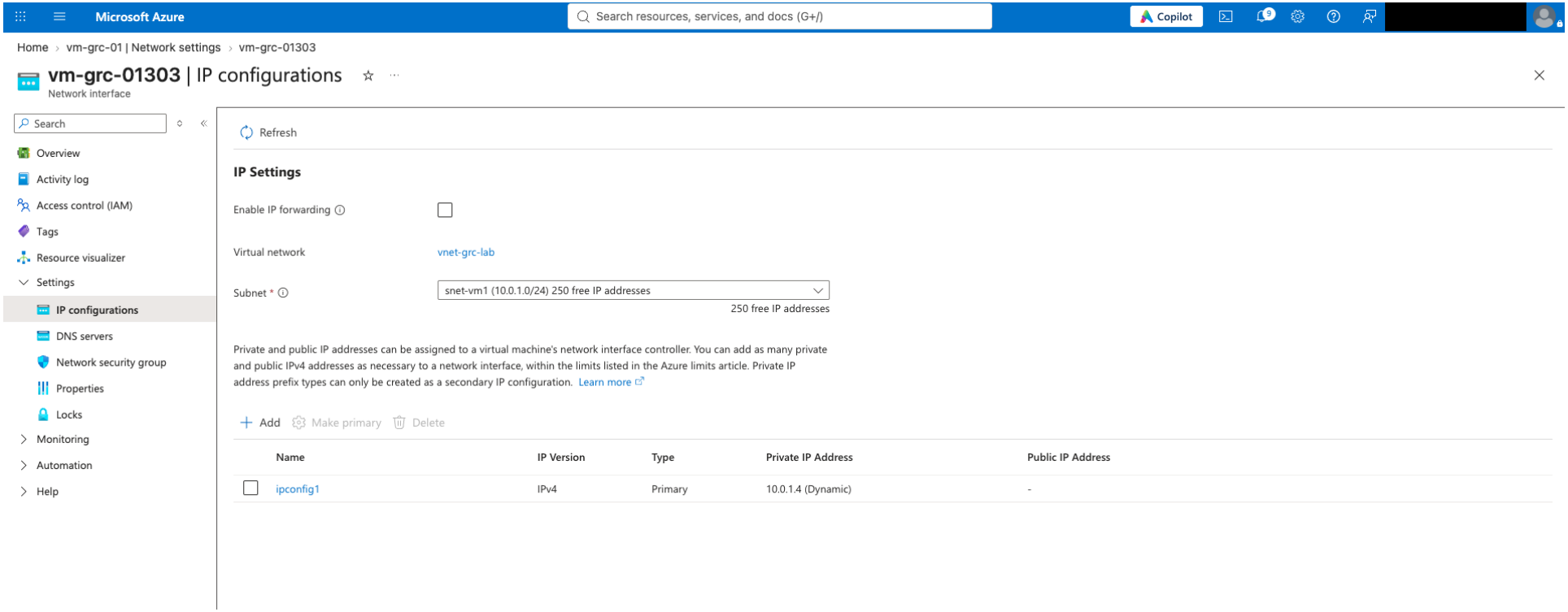
F01: POAM-01: Policy remediation tasks added managed identity and Guest Configuration extension. Controls: CM-6



F02: POAM-02 closed: unused AzureBastionSubnet removed. Controls: SC-7



F06: POAM-06: NIC IP forwarding disabled, no public IP. Controls: SC-7, AC-4



F09: POAM-09 closed: second budget alert recipient added. Controls: PM-3

